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**2026 5th i3s/Nobal Bannada ISS Online Class Component 09/02/2026:
- Sustainability in Building Conditioning Systems -**

Plan for today – learn more about the grid and about building energy systems!

- ⌘ Recap from yesterday/Outline for today**
- ⌘ Electrical power: Supply-side, Demand-side**
- ⌘ Heating and cooling systems – Intro**
- ⌘ Breakout room with exercise (home energy guesstimate)**
- ⌘ Heat pumps, water-based energy storage and its controls**
- ⌘ Wrap-up/Outlook**

Recap from yesterday – participant input!

- ⚙ Rating systems for buildings//human conscious -> make topics understandable/attractive to the broad majority of people
- ⚙ Refrigerant GWP – only considers refrigerant not system construction/lifecycle and energy emission effects
- ⚙ LEED provides a framework for measurable environmental sustainability, while WELL specifically centers on the physical, mental, and social well being of the building's occupant
- ⚙ Outlook for tomorrow -> Dr. Mansy: “Make buildings that are great for people”



... from the 2024 summer school

- Sustainable societies
- Life cycle assessments
- Biodiversity
- Global warming
- Sustainable refrigeration, air conditioning and heating
- Renewable energy
- Water management

- Electric vehicles
- Lithium extraction from geothermal water
- Hydrogen
- Energy storage
- Team building
- EU project SophiA
- EU project PrAEctiCe



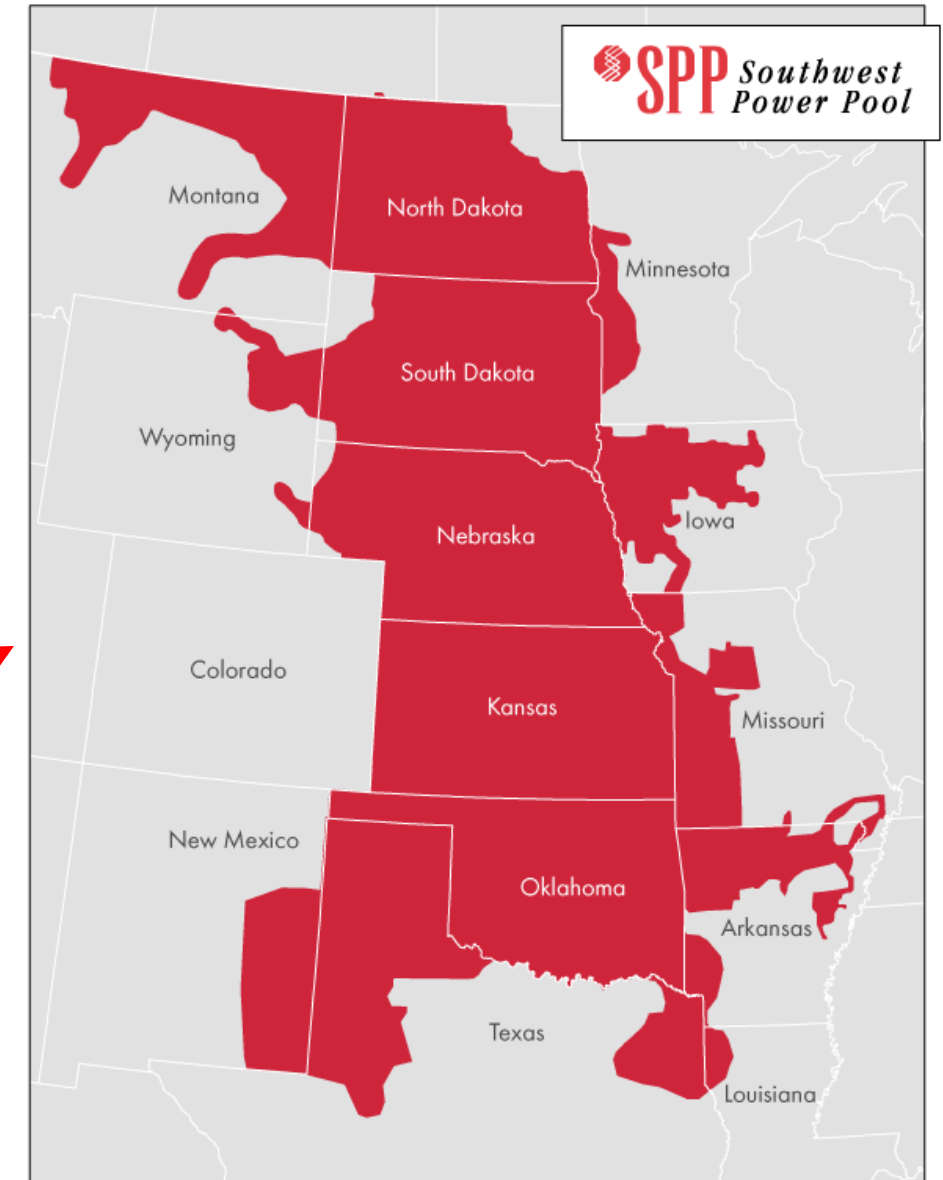
10 minutes: Electricity supply-side - Energy sources, storage, and environmental impacts: Objectives

⚙️ Introduce generation profile of various energy sources

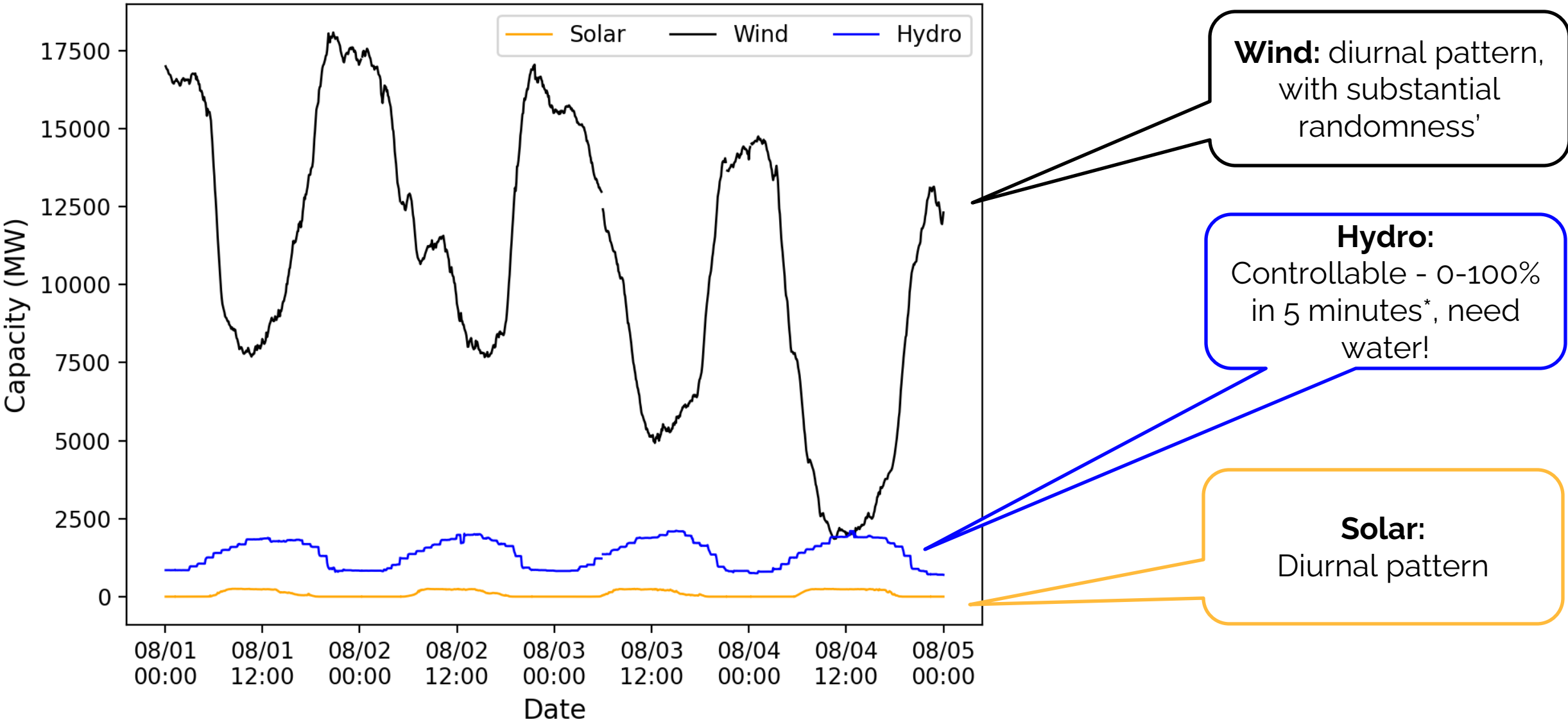
- Renewable (solar, wind, hydro)
- Fossil/dispatchable (nuclear, coal, gas)

⚙️ Introduce electricity storage options

- Geologically constrained vs. unconstrained
- Other parameters

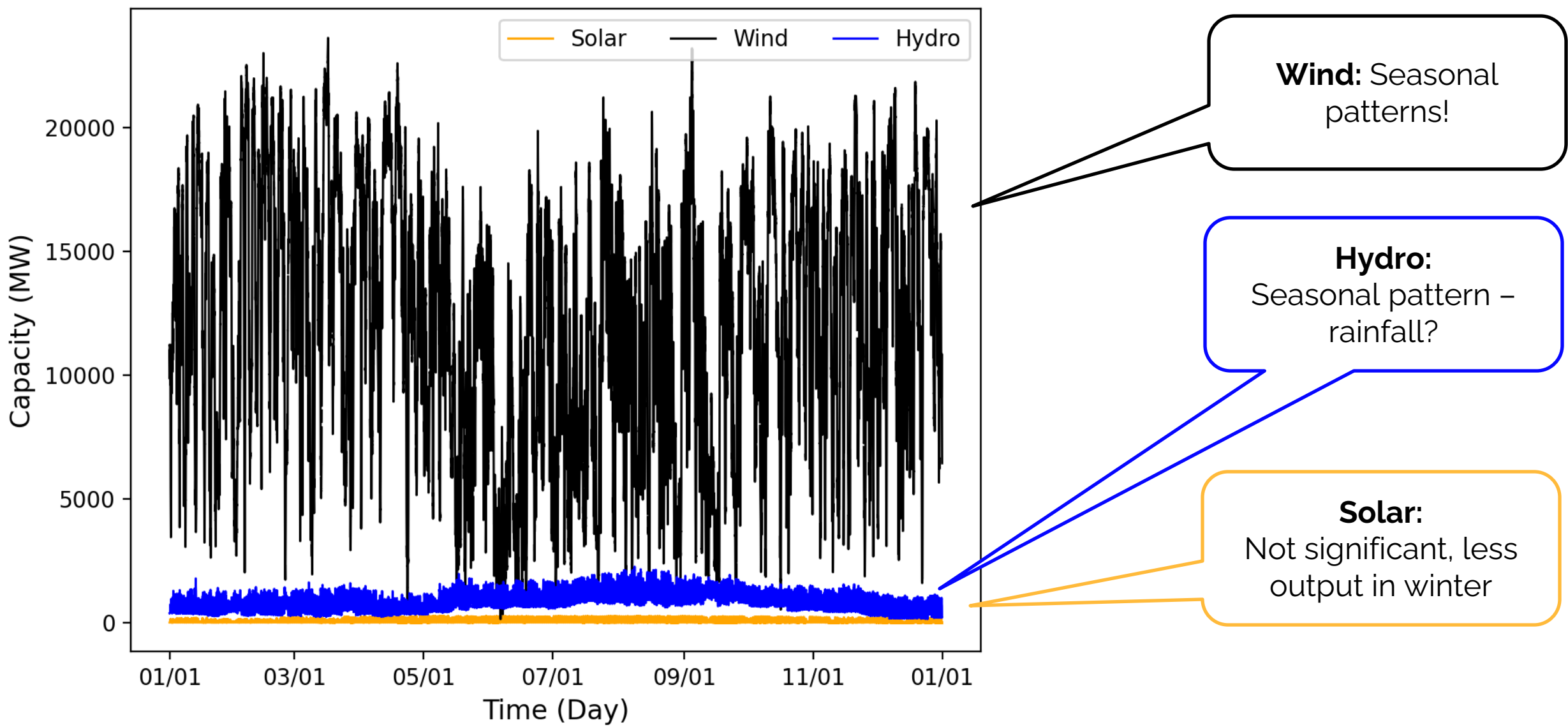


Load Generation Profiles – Renewables – 4 Days from 2023 (SPP)

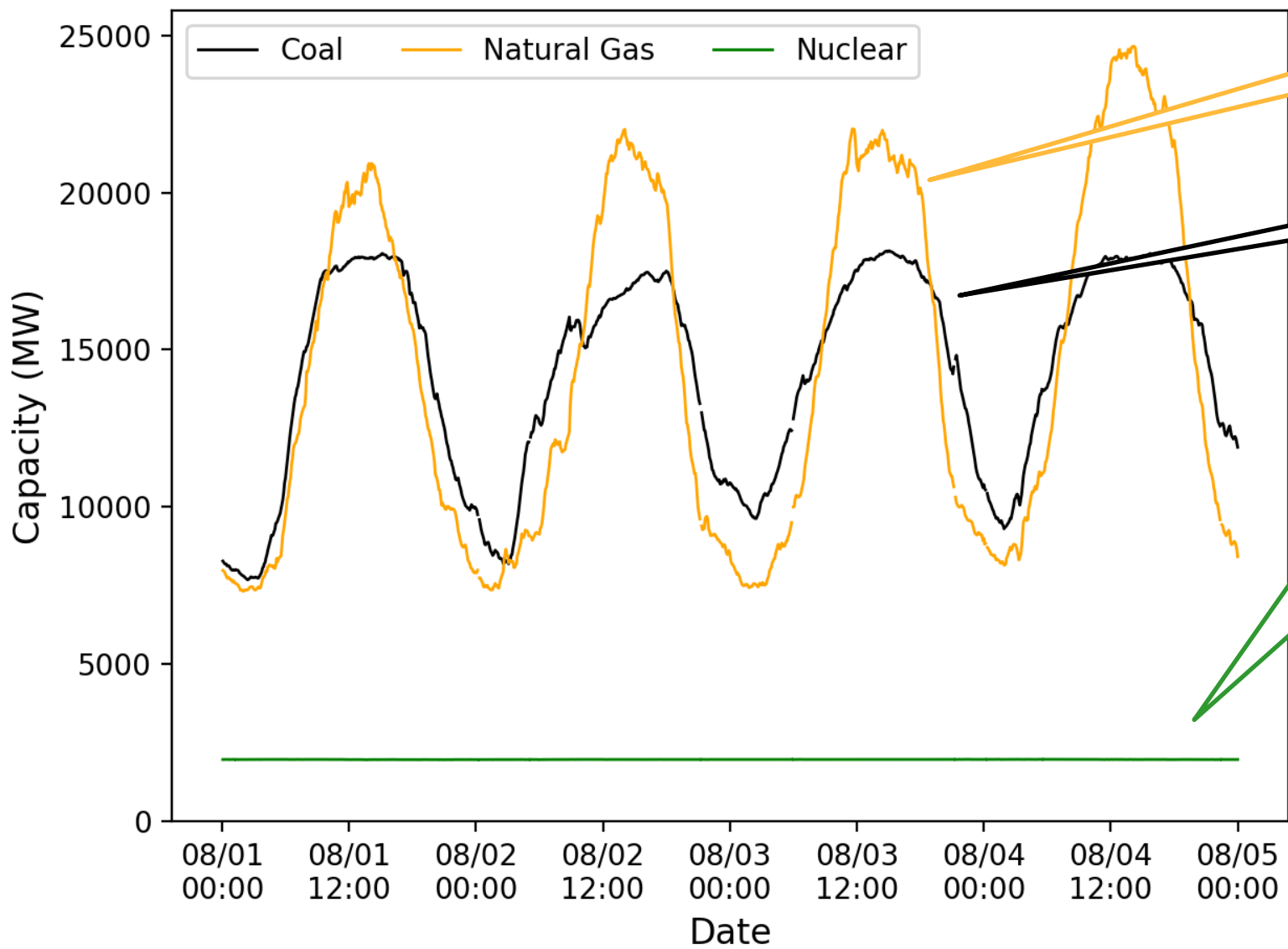


* <https://www.gevernova.com/gas-power/resources/articles/2017/load-following-power-plant>

Load Generation Profiles – Renewables – Full Year of 2023



Load Generation Profiles – Fossil Fuels: fill the gap!



Natural gas:
Diurnal pattern

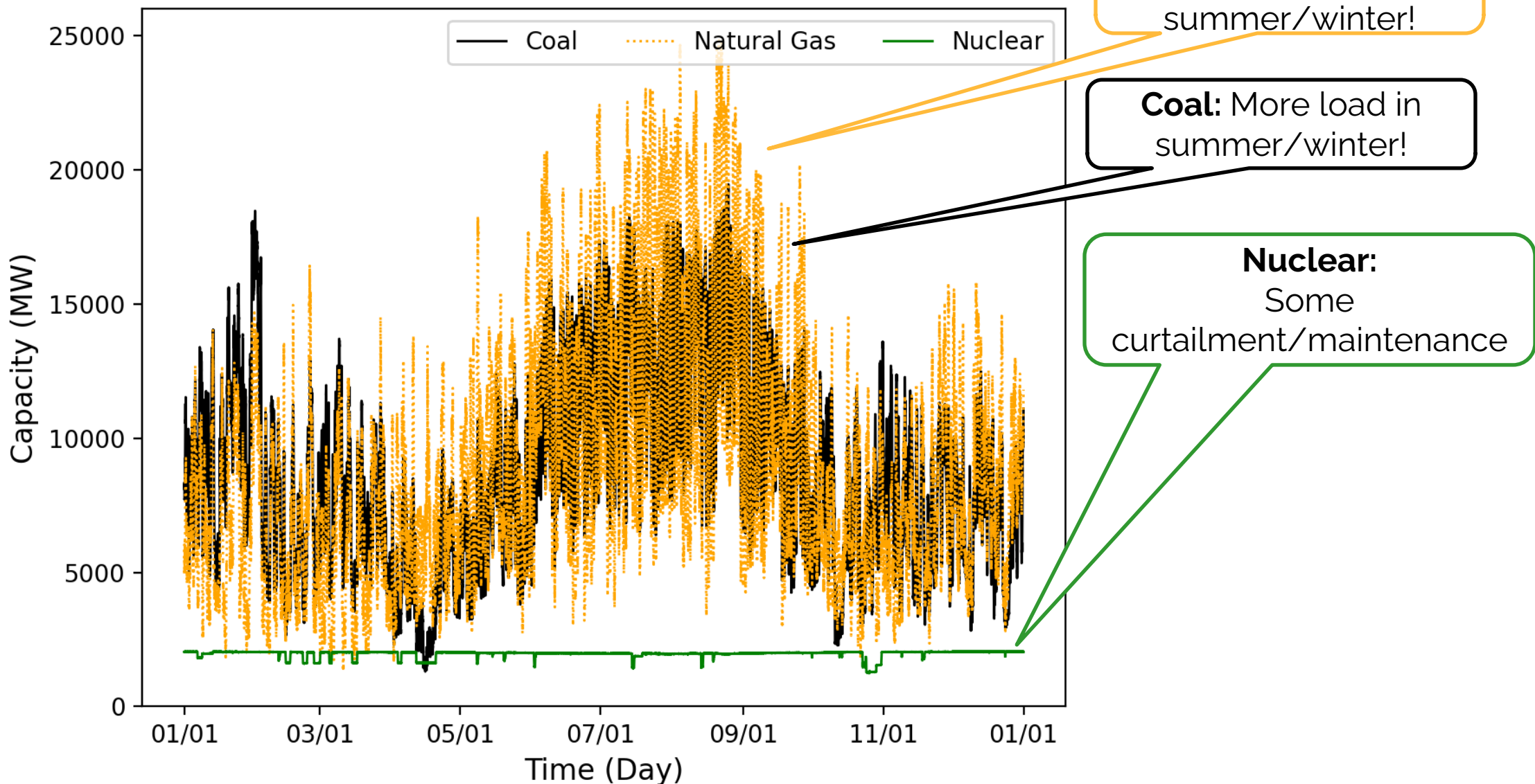
Coal: Diurnal pattern

Nuclear:
Baseload

Reaction times:
Gas as "peaker": 0-100% < 5 minutes*
Coal & Nuclear: 0-100% often >12 hrs**

* <https://www.gevernova.com/gas-power/resources/articles/2017/load-following-power-plant>
** <https://www.eia.gov/todayinenergy/detail.php?id=45956>

Load Generation Profiles – Fossil Fuels: fill the gap!



10 minutes: Demand-side – Load profiles

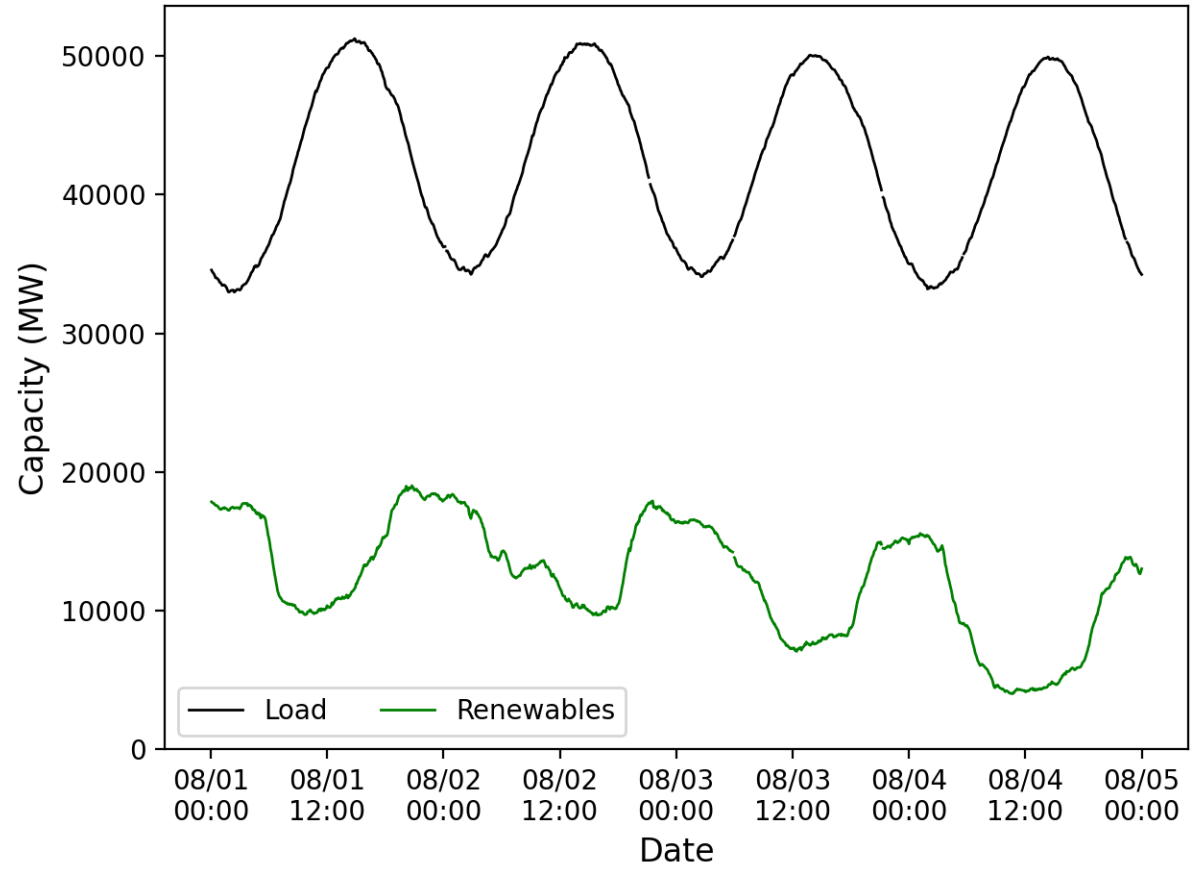
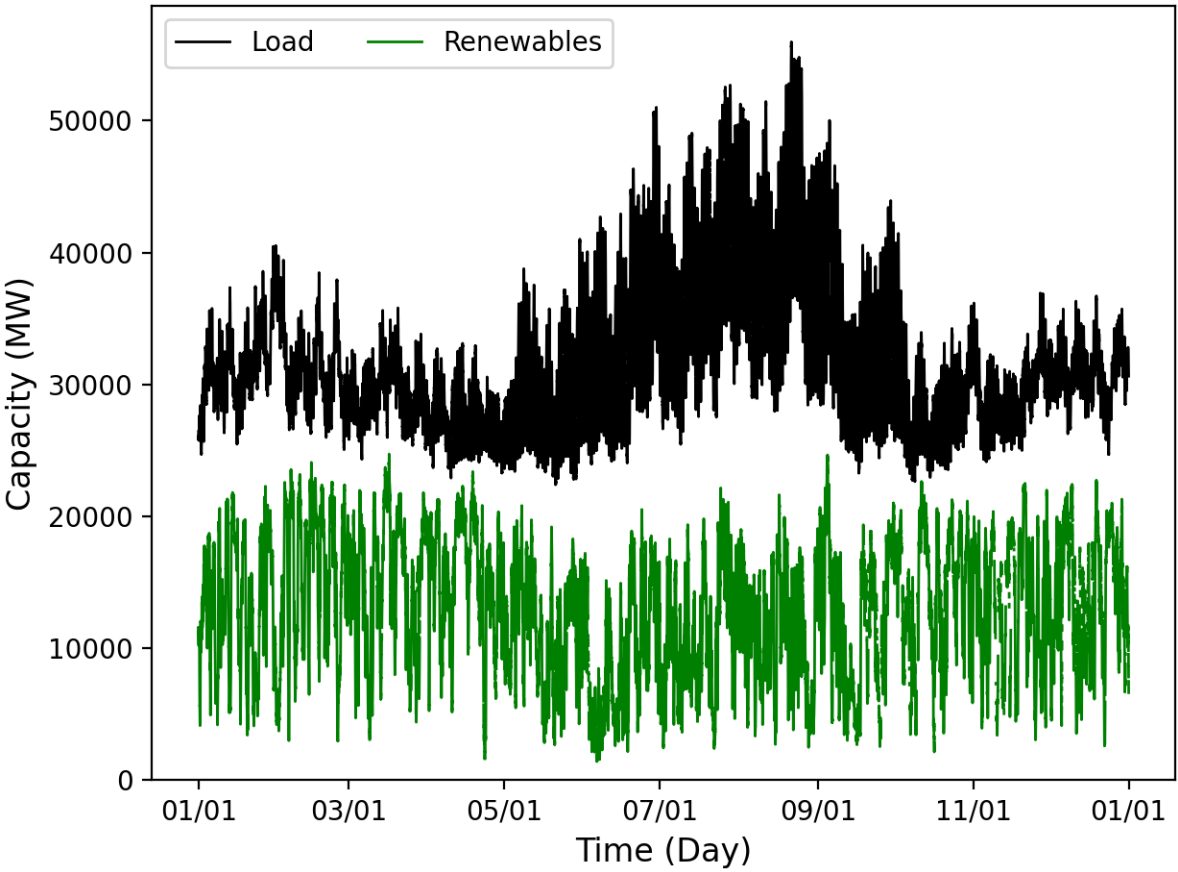
Objectives

- ⚙ **Introduce typical US load profiles**
- ⚙ **Introduce Duck/alligator curve**
- ⚙ **Fast forward DE/Europe – will have to adopt AC**

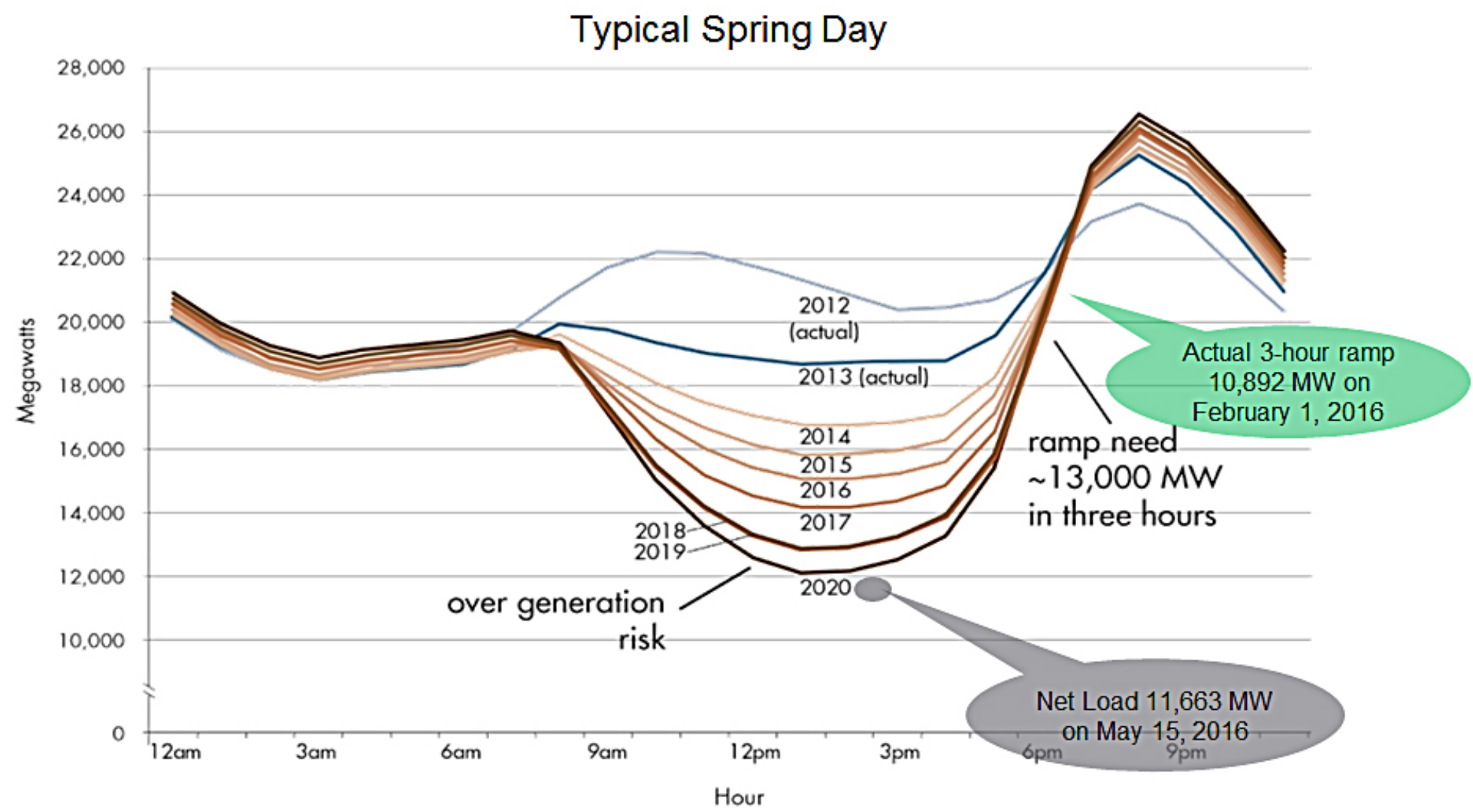
“Typical” US electricity load profile

- ⚙ **Driven by AC/HP loads, water heating, and commercial/industrial loads**
 - US has a wide range of climates
 - US has locally more extreme short time weather changes than Europe
 - US has a wide range of regional industries
- ⚙ **SPP data example next slide!**

SPP Load Data Examples (2023) – Renewables don't track well!

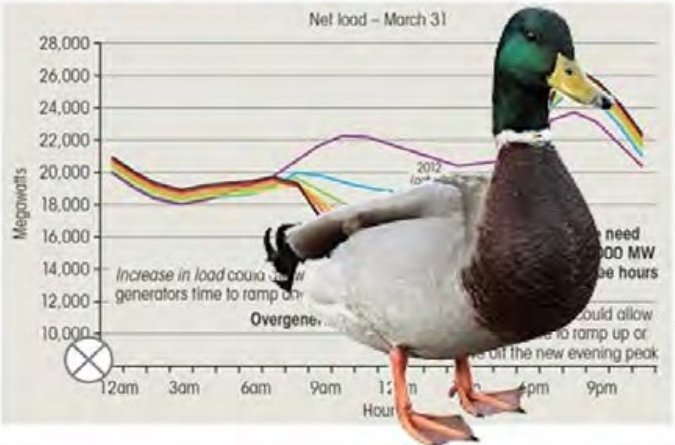


Net load on grid after consideration of solar-heavy renewable electricity in California.

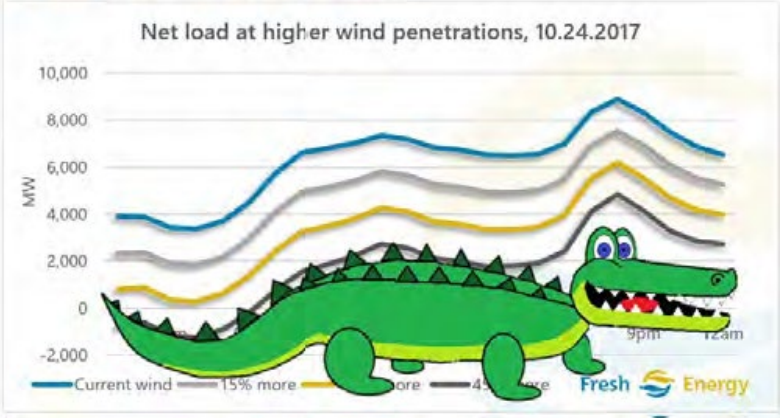


Grid net load post renewables get more and more extreme – animals to the help!

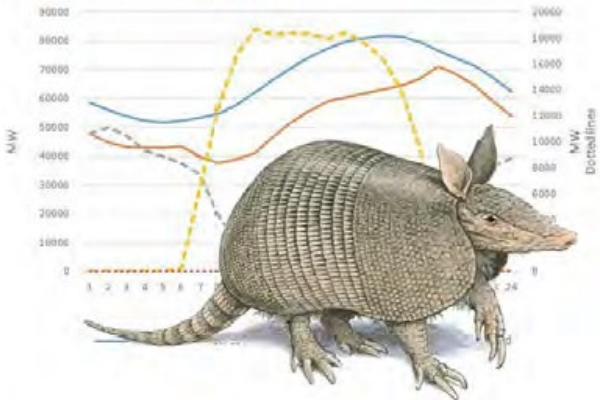
California: The Duck Curve



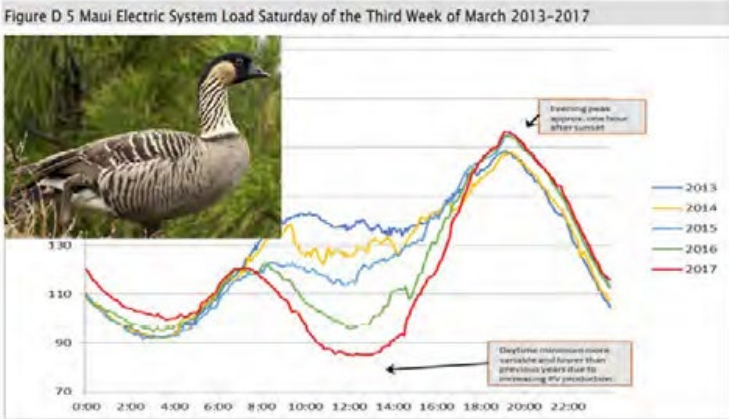
Midwest: The Gator Curve



Texas: The Armadillo Curve

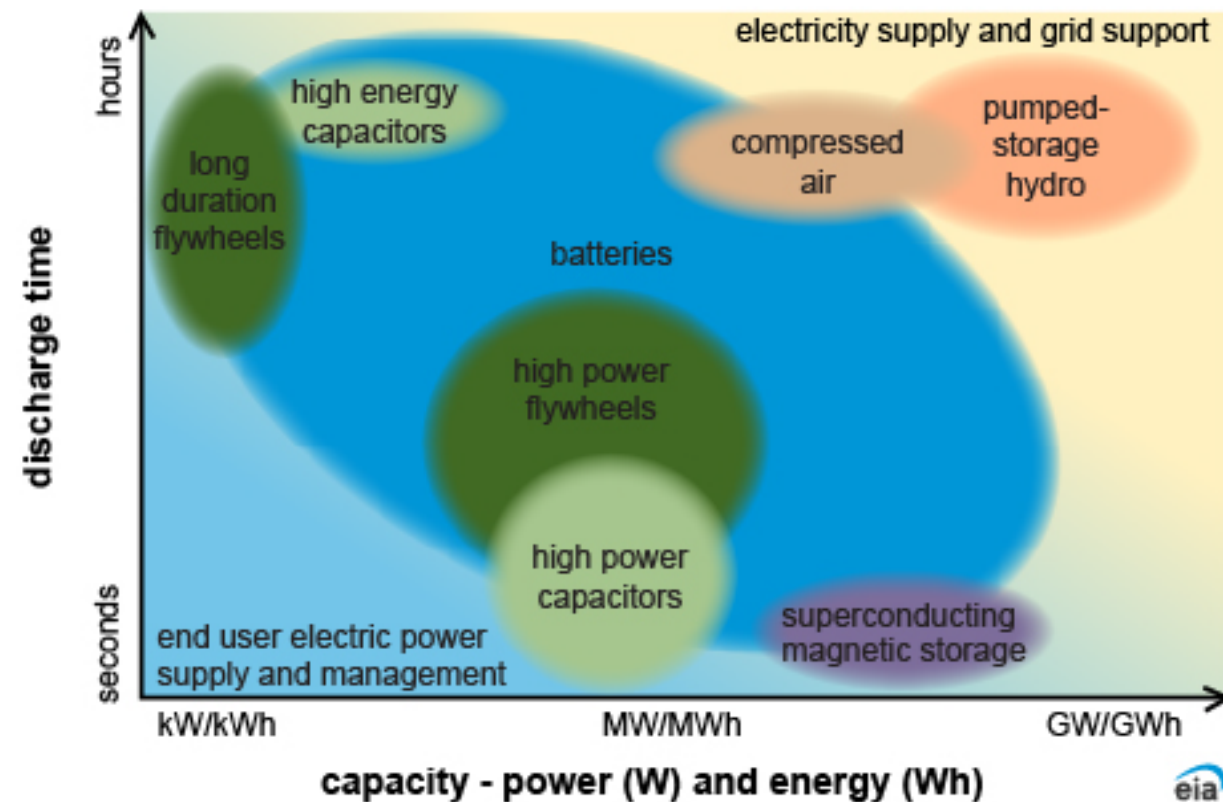


Hawaii: The Nene Curve



Electricity Storage Options

- ⚙️ **Discharge power and capacity**
 - Long term vs. short term support
- ⚙️ **Geologically constraint storages**
 - Pumped storage
 - Compressed air
- ⚙️ **Environmentally impact**
 - All, especially batteries (mining/production)
 - Round trip efficiency (40...90+%; McCracken, 2010)
- ⚙️ **Other parameters**
 - Capital cost
 - Life expectancy



10 minutes: Heating and cooling systems – intro

Objectives

⚙ **US – Air based systems**

- Air distribution or window AC or mini-split
- AC/HP
- Gas furnace
- Electric heat

⚙ **Germany – Water based systems**

- Single-room radiators
- Gas/oil boiler
- Heat pump
- Electric heat with storage brick (legacy?)

How an Air Conditioner Works:

Similar to how a refrigerator works, air conditioners transfer heat from a home's interior to the warm outside environment.

A Evaporator

Cooling coils remove heat and humidity from the air using refrigerant.

B Blower

A blower (or fan) circulates air over the evaporator, dispersing the chilled air.

C Condenser

Hot coils release the collected heat into the outside air.

D Compressor

A pump that moves refrigerant between the evaporator and the condenser to chill the indoor air.

E Fan

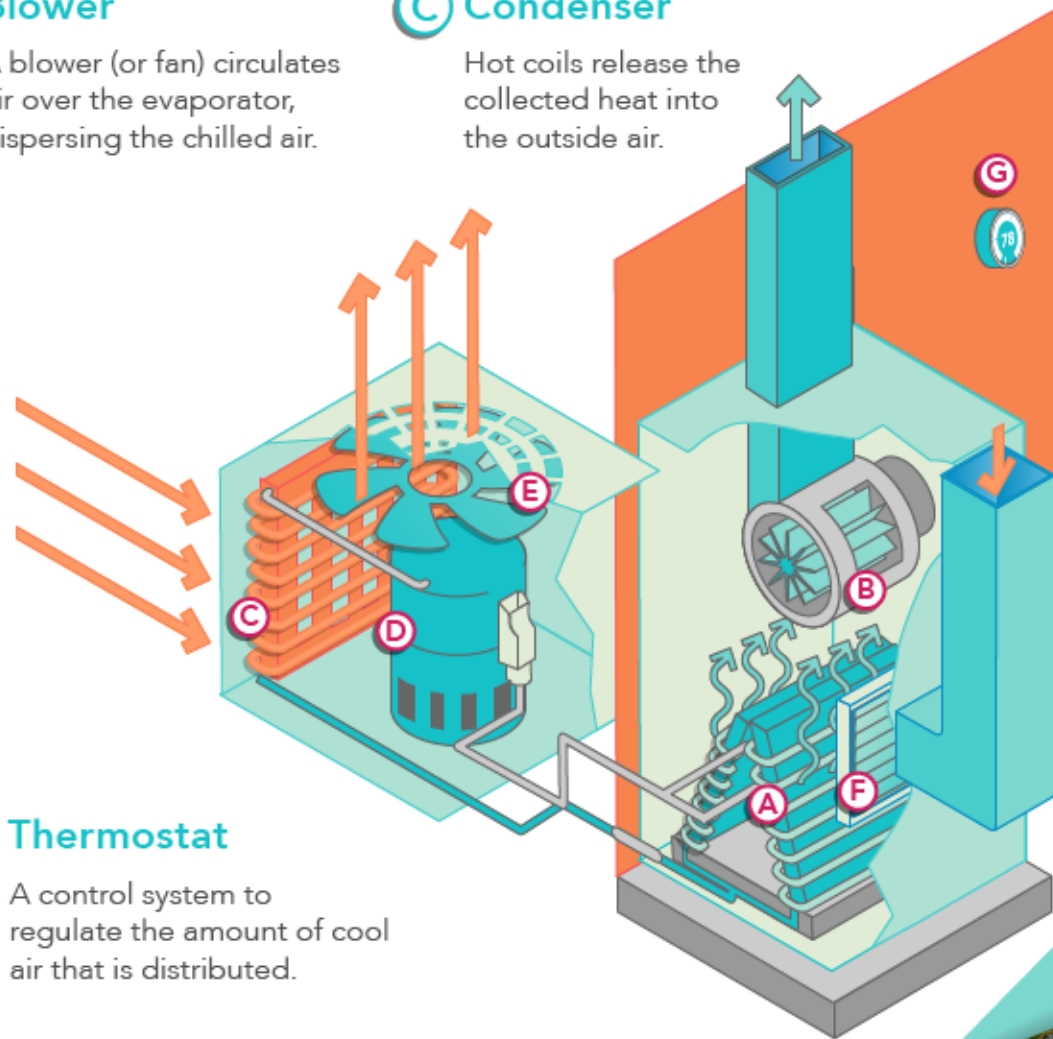
A fan blows air over the condenser to dissipate the heat outside.

F Filter

Located in the air conditioning unit to remove particles from the air.

G Thermostat

A control system to regulate the amount of cool air that is distributed.



Fast forward DE/Europe – will have to adopt AC!

⚙️ **Bad news:**

- Will need AC at some point

⚙️ **Good news:**

- Easier to deal with warmer winters
- Synergies: If need AC, may as well use a heat pump!

History:

All ▾

Show non-metric

More



20 minutes: Breakout room with exercise (carbon footprint guesstimate)

Objectives

⚙️ Estimate your carbon footprint

- Option for everyone: <https://www.carbonfootprint.com/sustraxvita.html#calculator>
- Germany:
 - https://www.naturefund.de/wissen/co2_rechner/
 - https://uba.co2-rechner.de/de_DE/#calculator
- US: [What is your carbon footprint? | Carbon Footprint Calculator \(nature.org\)](#)

⚙️ Compare your carbon footprints within your group

- What produces most emissions?
- Where can you improve?

⚙️ How would your footprint change if you switched to a heat pump?

10 minutes: Report back from breakout rooms

Objectives

⚙ **Compare carbon footprints (annual)**

- 47 tons/year (Christian + wife + son)
- 24,000 kg (2025 i3s - Christian – Germany, now in US)
- 15,000 kg (2025 i3s - Michael – Germany/Switzerland – need to plant 40 trees, not accounting hardware)
- 11 tons/year (Jung) – “not accurate”
- 9.8 tons is German average as per Umweltbundesamt (Federal environmental agency)
- 9.4 tons (San)
- 5.4 tons (Farnoosh)
- 4 tons/year (Robin) – German calculator not that detailed, didn't include public transport
- 4 tons/year (Aaron) – difficult to estimate heating fuel consumption
- 8 tons/year (Robin #2 calculation)
- 3.5 to 4 tons (Chat GPT) – WWF: 9 tons/year (program/assumptions/level of detail)
- 3.4 tons (San #2)
- 3 tons (Meinel)
- 1,178 kg (2025 i3s - Olivia – needs to plant one tree, Switzerland)

10 minutes: Report back from breakout rooms

Objectives

✿ Discussions

- Uncertainty on values, difference between calculators; difficult to estimate inputs
- Optimal time of replacement against higher performing/more efficient products
- Bike vs. car
- Greenwashing on cars - rating system based on fuel/weight of car?!?
- How much is car actually used vs. standing around?
 - Car sharing – Karlsruhe: Stadtmobil, similar in Switzerland and US
- Maintenance – malfunctioning heating radiator

✿ How can we improve?

- Christian - Cut back on airplane travel, cut back on water consumption, get a higher efficiency AC//new windows
- San – transportation in summer – bike, less car, cut back on meat and products.
- Carbon offsets? there has been a very interesting documentation about this in german television. From what I remember, it's main points were that the compensation leads to planting tree monocultures and the forced moving of local people to make space for the monocultures.
- Jung – flights are a large share of emissions, in other areas could try to cut back but difficult

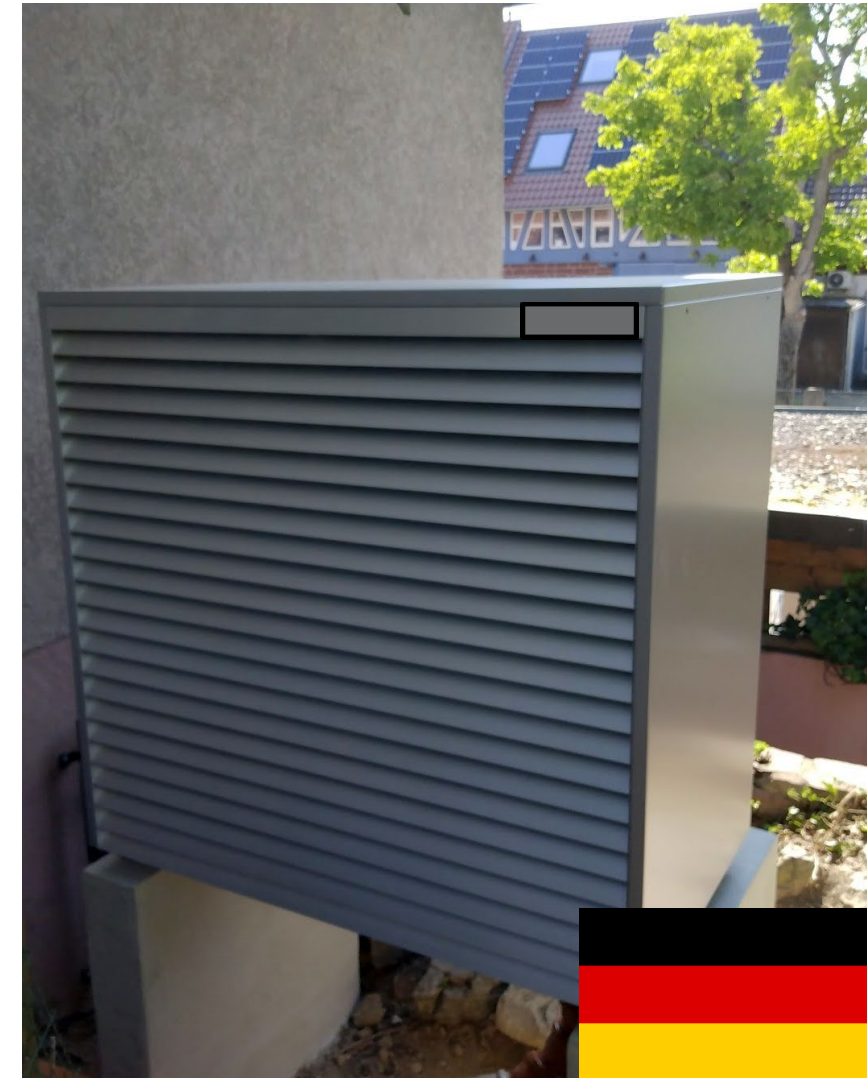
15 minutes: Heat pumps and their challenges

Objectives (system age at time of picture)

- ✿ General intro heat pumps
- ✿ Heat pump vs. building heating requirement
- ✿ Heat pump frosting
- ✿ Revised heat pump vs. building heating requirement



Air-Air Split System Heat Pump,
Stillwater, Oklahoma (~10Y old)



Air-Water Split System Heat Pump,
Neulingen, Germany (new)

Heat pumps are more efficient than electric resistive heat and can allow cooling of buildings but have some drawbacks.

⚙ **Use Electricity to “pump heat” up in temperature**

- Heating/HP mode: Pump heat from a source (outdoors, ground) into a building
- Cooling/AC mode: Pump heat from the inside of the building to the outside
- Hybrid models including water heating: Heat domestic water and building
- Heat pump water heaters: Pump heat to heat water ($\sim 45^{\circ}\text{C}$)
- Industrial heat pump/steam generation: Pump heat to high temperatures

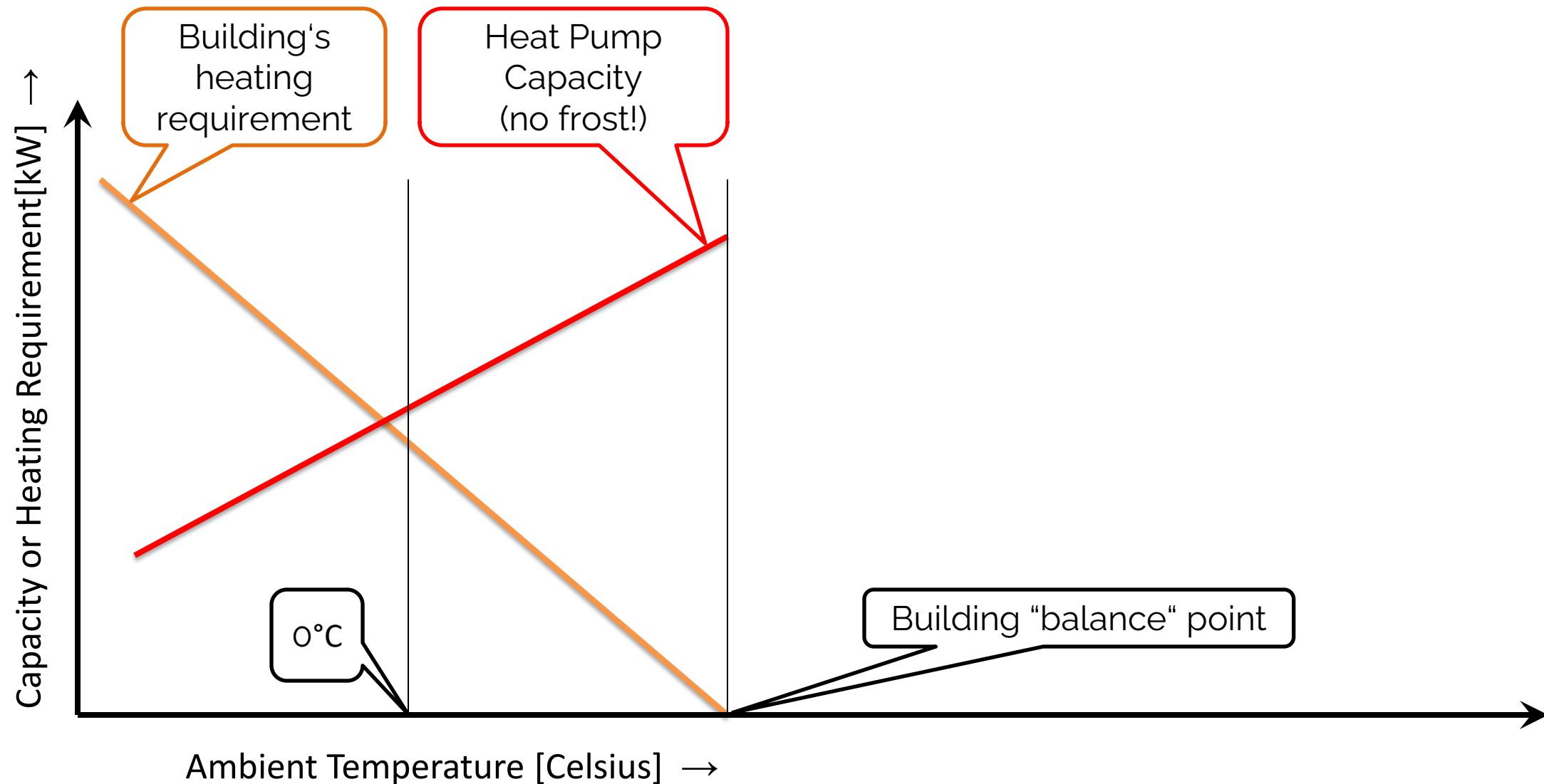
⚙ **Promise of heat pumps**

- Heating: Use less electricity than electric resistance heating (COP 1 vs COP >3)
- Cooling: No longer need to harvest ice in winter to use in summer

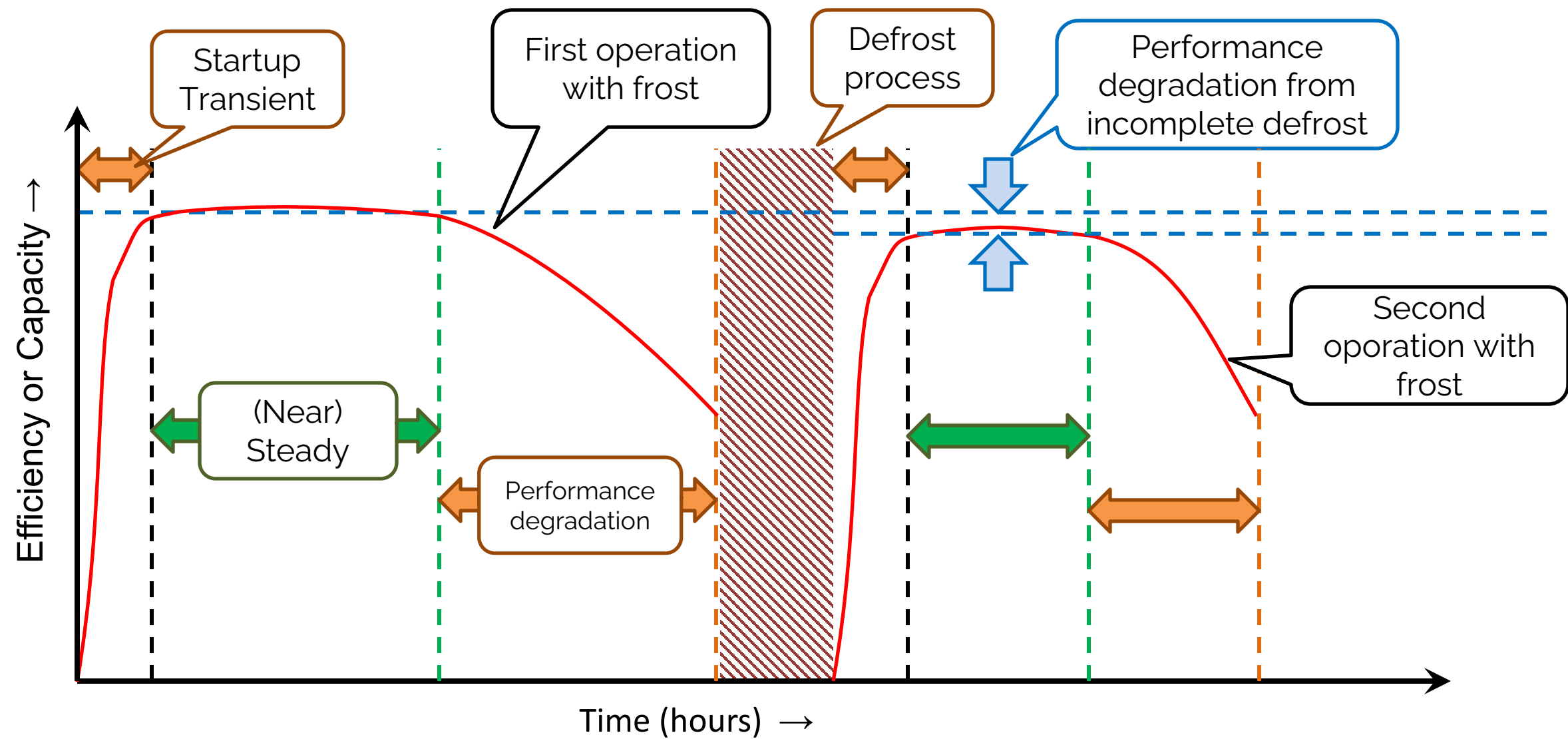
⚙ **Issues**

- Complex devices – maintenance cost
- Initial cost, operating cost if gas is cheap
- Dealing with frosting

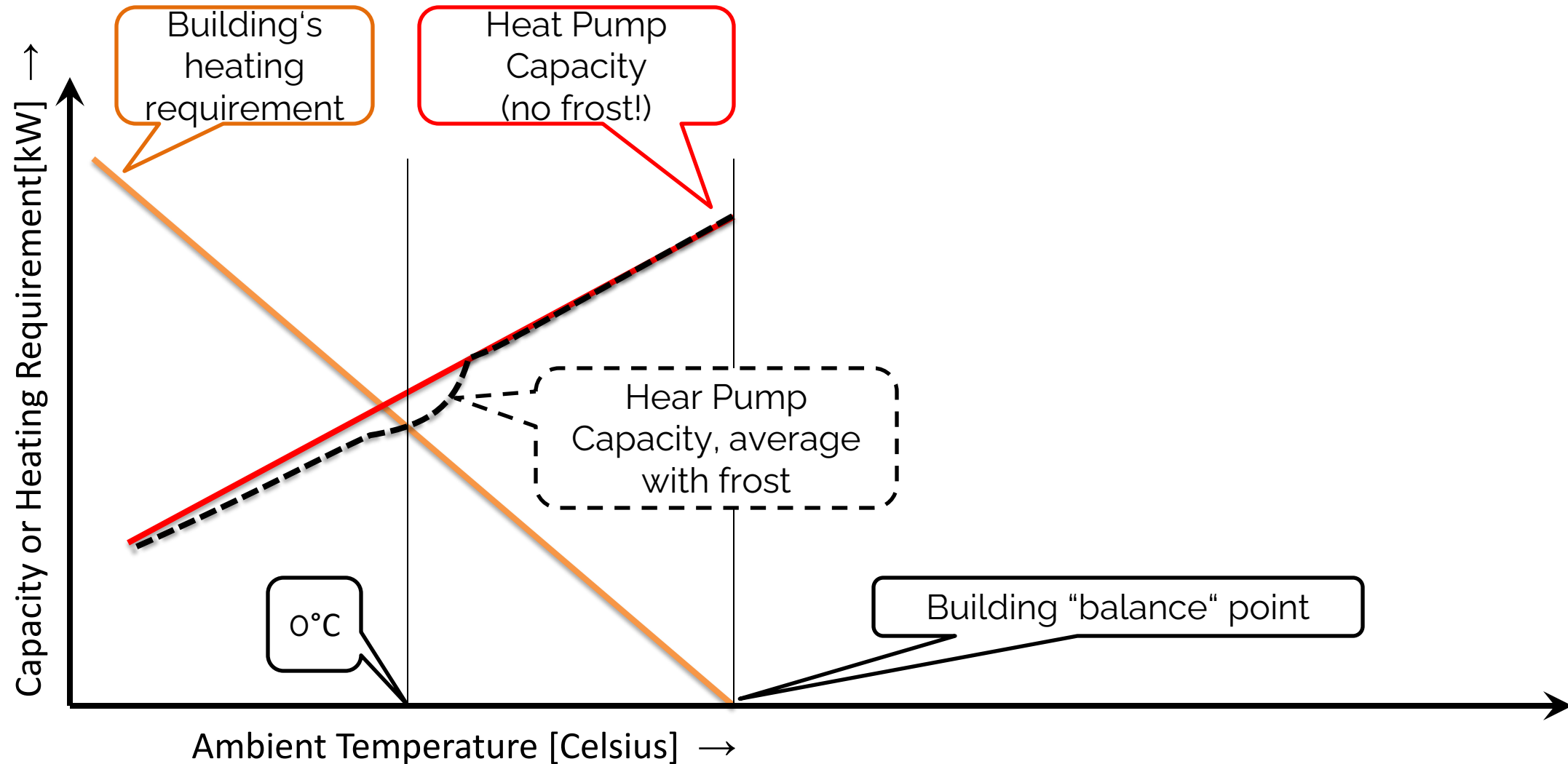
Heat pump capacity is larger at high temperatures – opposite to what the building needs.



Heat pumps need to be defrosted which reduces efficiency and capacity



Heat pump capacity is larger at high temperatures – opposite to what the building needs. **Frosting further contributes to the issue.**



5 minute Q&A

Objectives

- ⚙ Have students ask for clarification/identify gaps**
- ⚙ Ask questions/foster discussions if no questions from audience**

15 minutes: Water based thermal energy storage intro

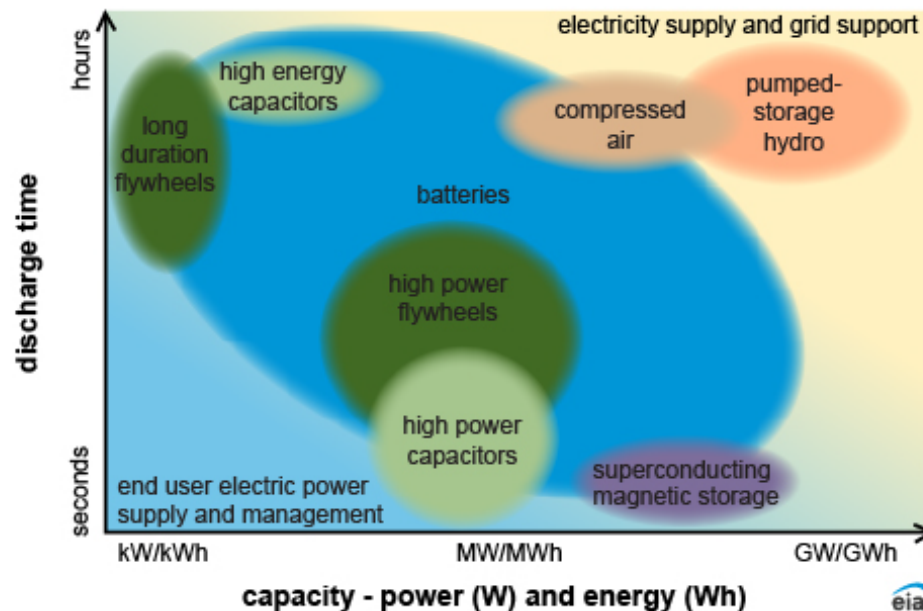
Objectives

- ⌘ General intro to TES (why/what for)**
- ⌘ Water based TES – TriCoil vs. fully hydronic**
- ⌘ How large of a storage is needed?**

Recap – Energy Storage – there were some issues!

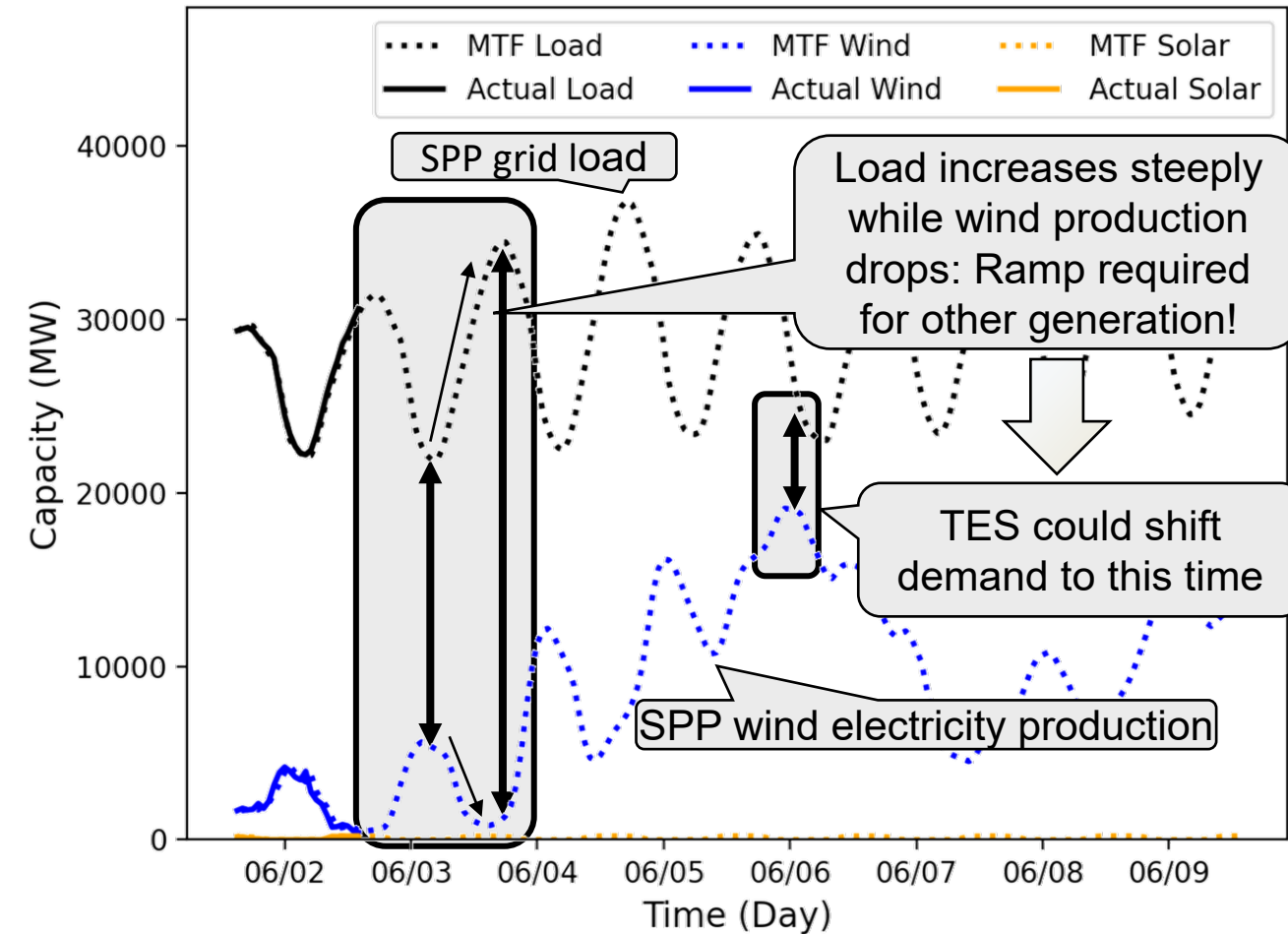
Electricity Storage Options

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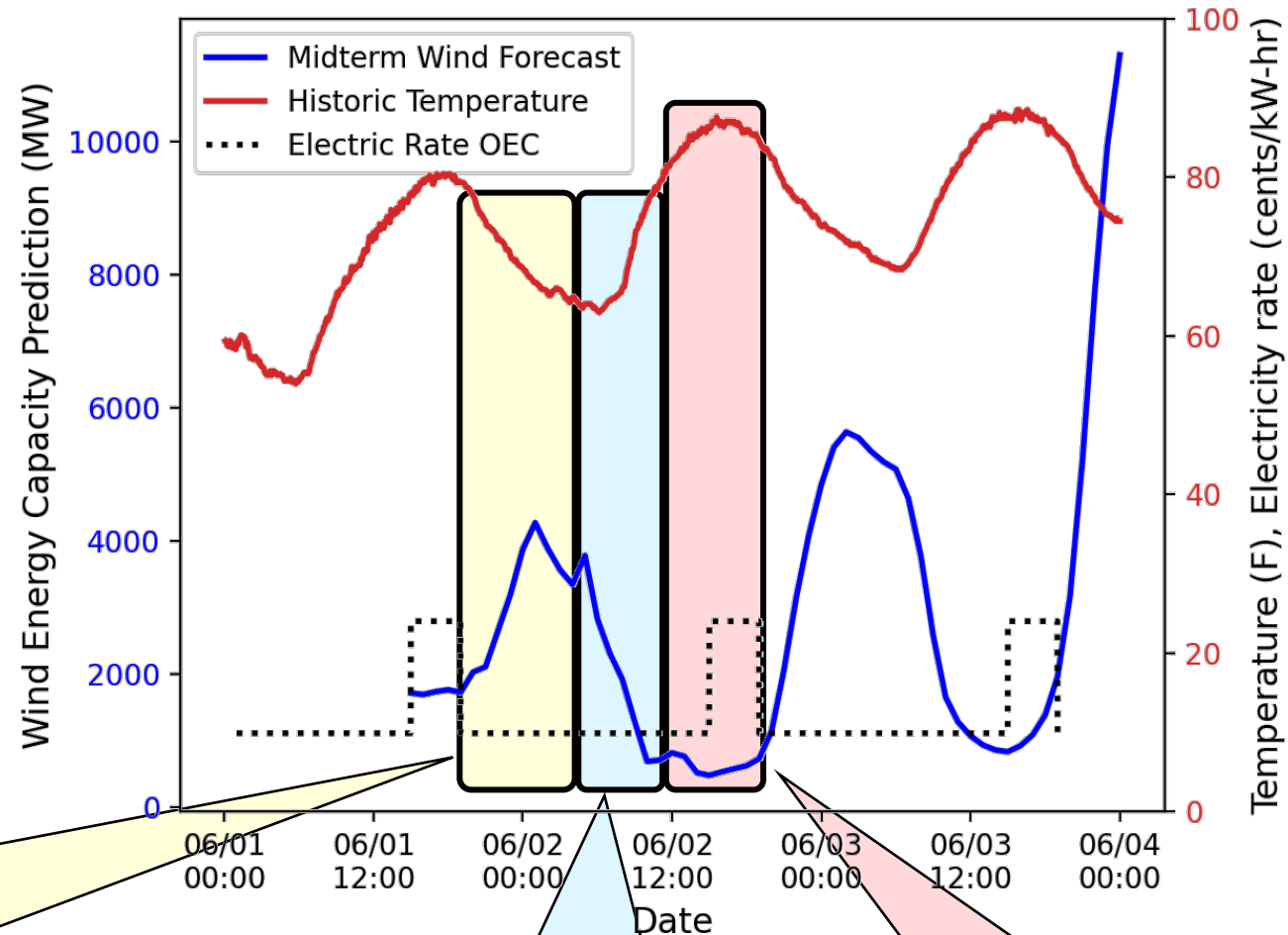
Renewables not always available. TES to help handle variations in renewables!

- ❁ **SPP: 68.78% wind energy penetration peak (2019)**
- ❁ **Almost 7x increase in capacity between 2009 and 2019**
- ❁ **Wind doesn't always blow!**
 - “Ramps” of steep in/decrease issue for coal and nuclear
 - Need to shift!
- ❁ **Thermal storage**
 - Store hot/cold
 - Decouple majority of electricity consumption from when cooling/heating is needed on a diurnal scale



Midterm Forecast (MTF) and actual for electric load and generation for the Southwest Power Pool (SPP). Data from SPP (2021).

Thermal energy storage – charge when abundant electricity, discharge when lack of (cheap renewable) electricity.



Maximize consumption (charging TES & cooling) to reduce electricity consumption ramp and maximize wind energy use

Discontinue TES charging to reduce ramp for conventional power plants

Discharge TES, reduce demand on grid during low wind production

Thermal Energy Storage Options for Space Conditioning

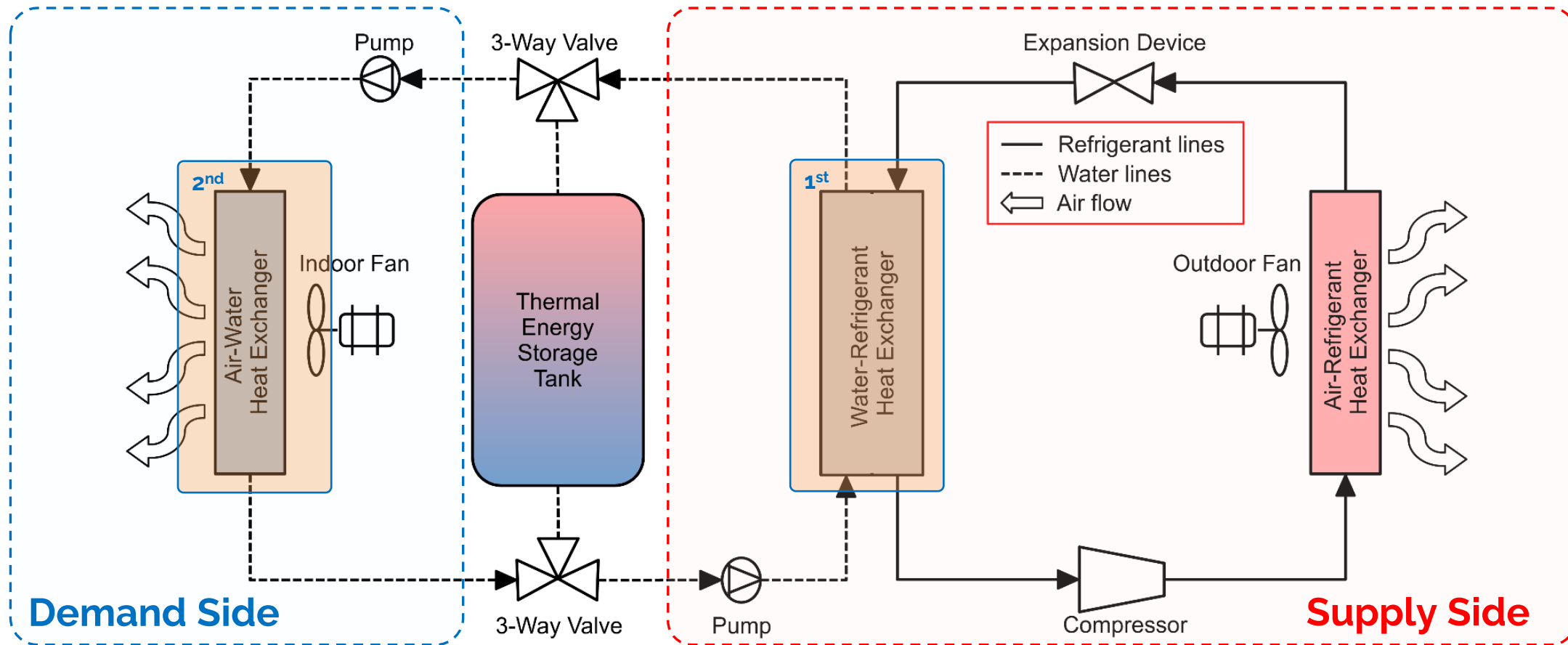
- ✿ **Sensible storage:** Change in storage medium's temperature as storage
 - “Heavy” buildings' thermal mass
 - Thermally activated building systems (HS Biberach, DE)
 - Environmentally benign water in dedicated tanks
- ✿ **Latent or “Phase Change” storage:** Liquid -> Solid
 - Water/Ice in dedicated tanks
 - Paraffins and other 'PCMs' in building or dedicated tanks
- ✿ **Thermochemical:** Chemical reaction
 - Not yet mainstream



Real frogs like water!

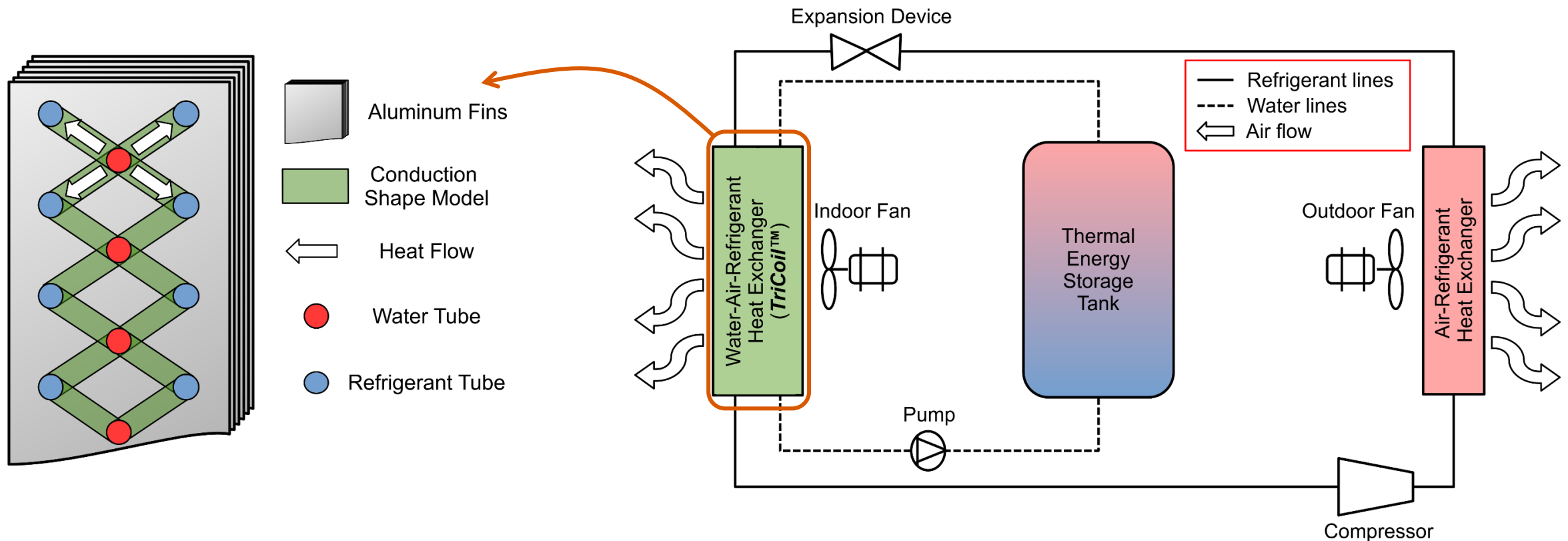
US: Integration of water-based TES as fully hydronic system!

- ❁ **Fully hydronic system:** two heat transfer processes on the use-side
- ❁ **Can apply same concept in European radiator systems for heating mode**



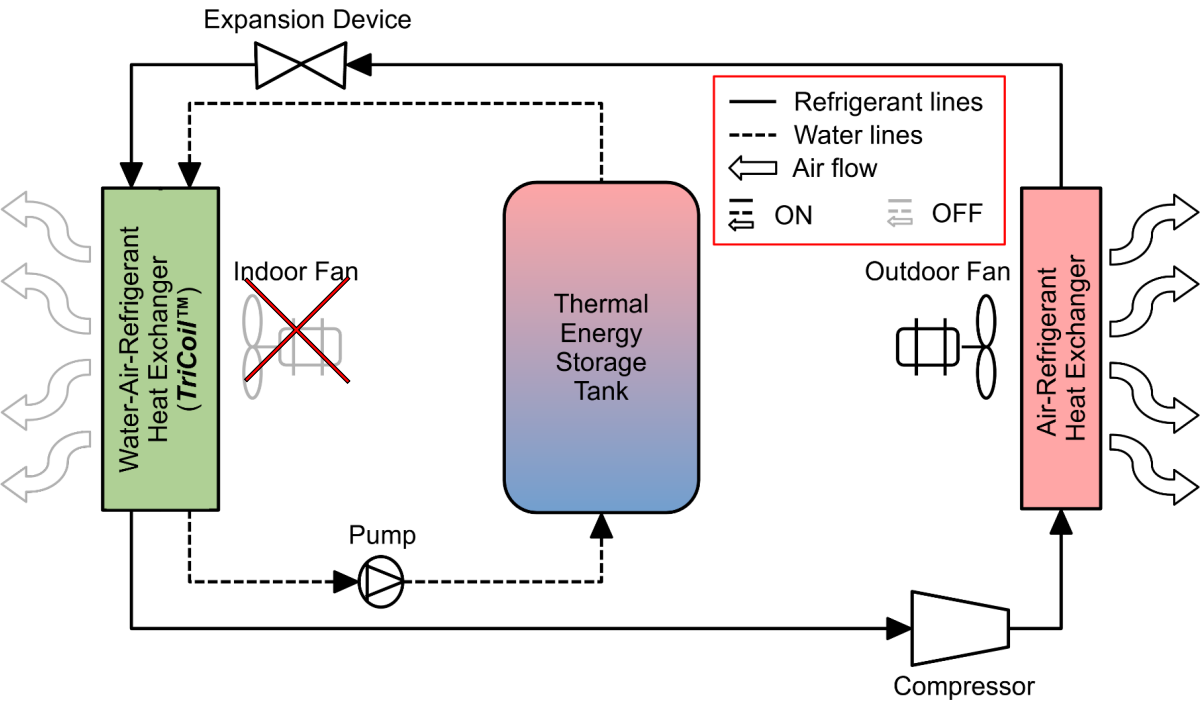
US: TriCoil™ for air-based systems integrates “into” indoor heat exchanger.

- ✿ Combine efficiency of conventional split system with TES ability
- ✿ TriCoil is a fin-tube heat exchanger using cross-fin conduction for charging
- ✿ Operate as “normal” heat exchanger in discharging and direct operation modes

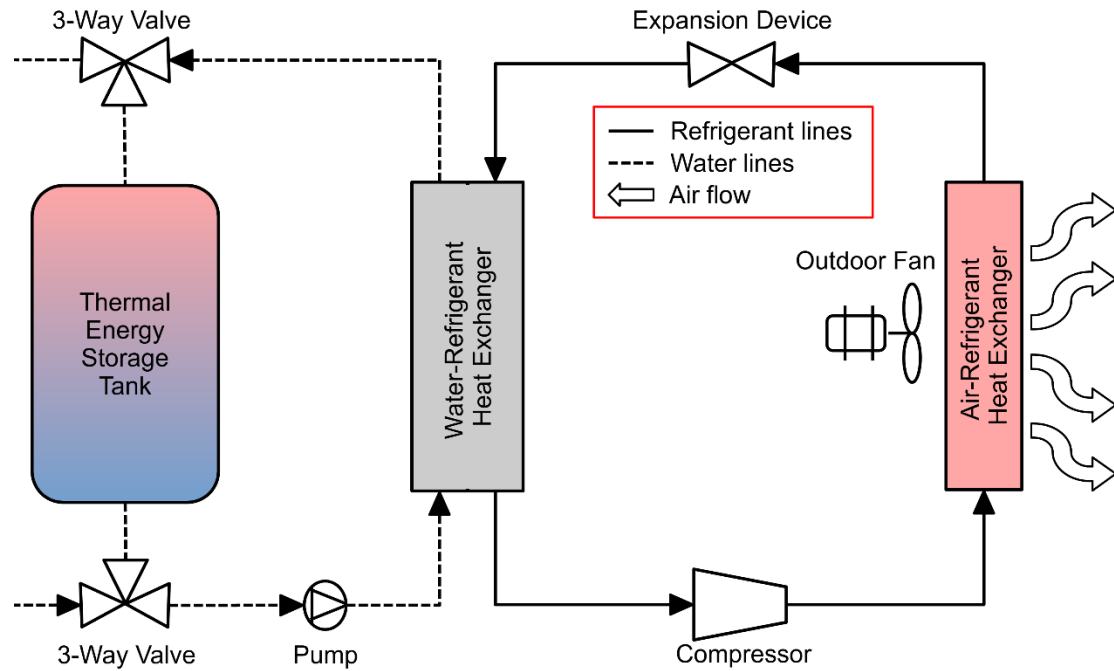


TriCoil™ in charging mode acts similar to a hydronic system.

Cooling mode shown in this and next few slides.

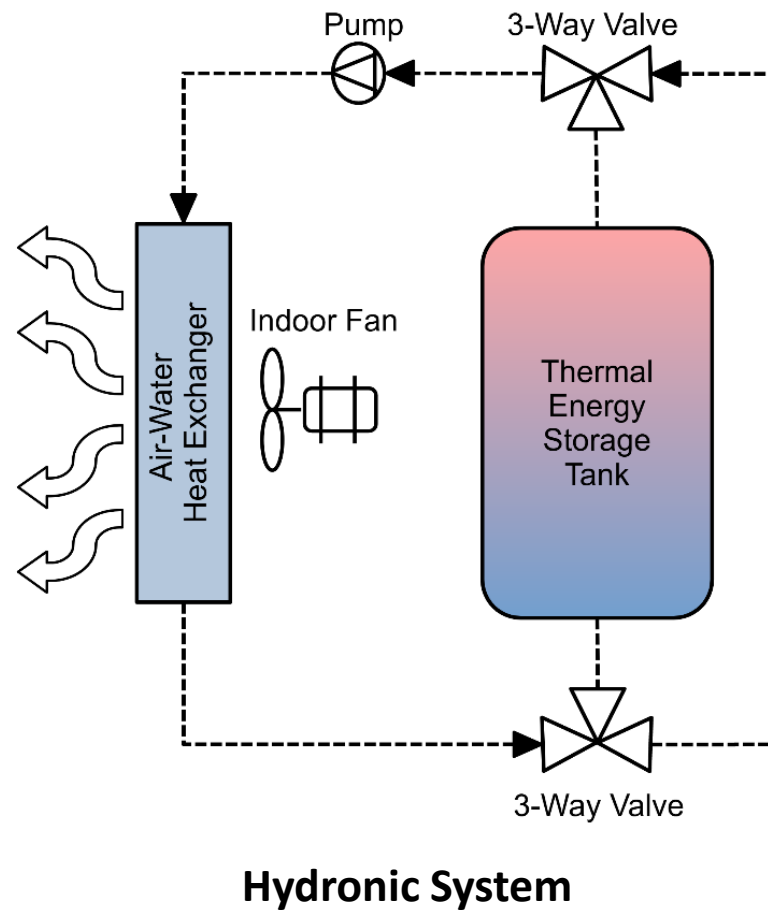
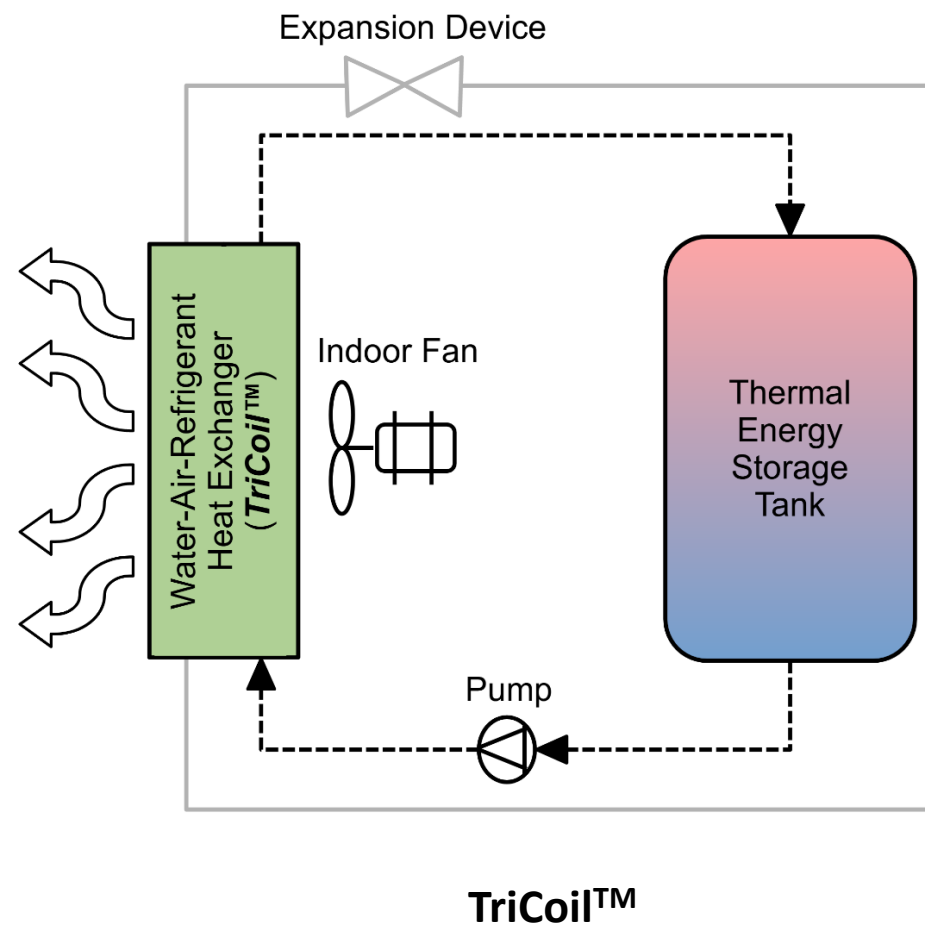


TriCoil™

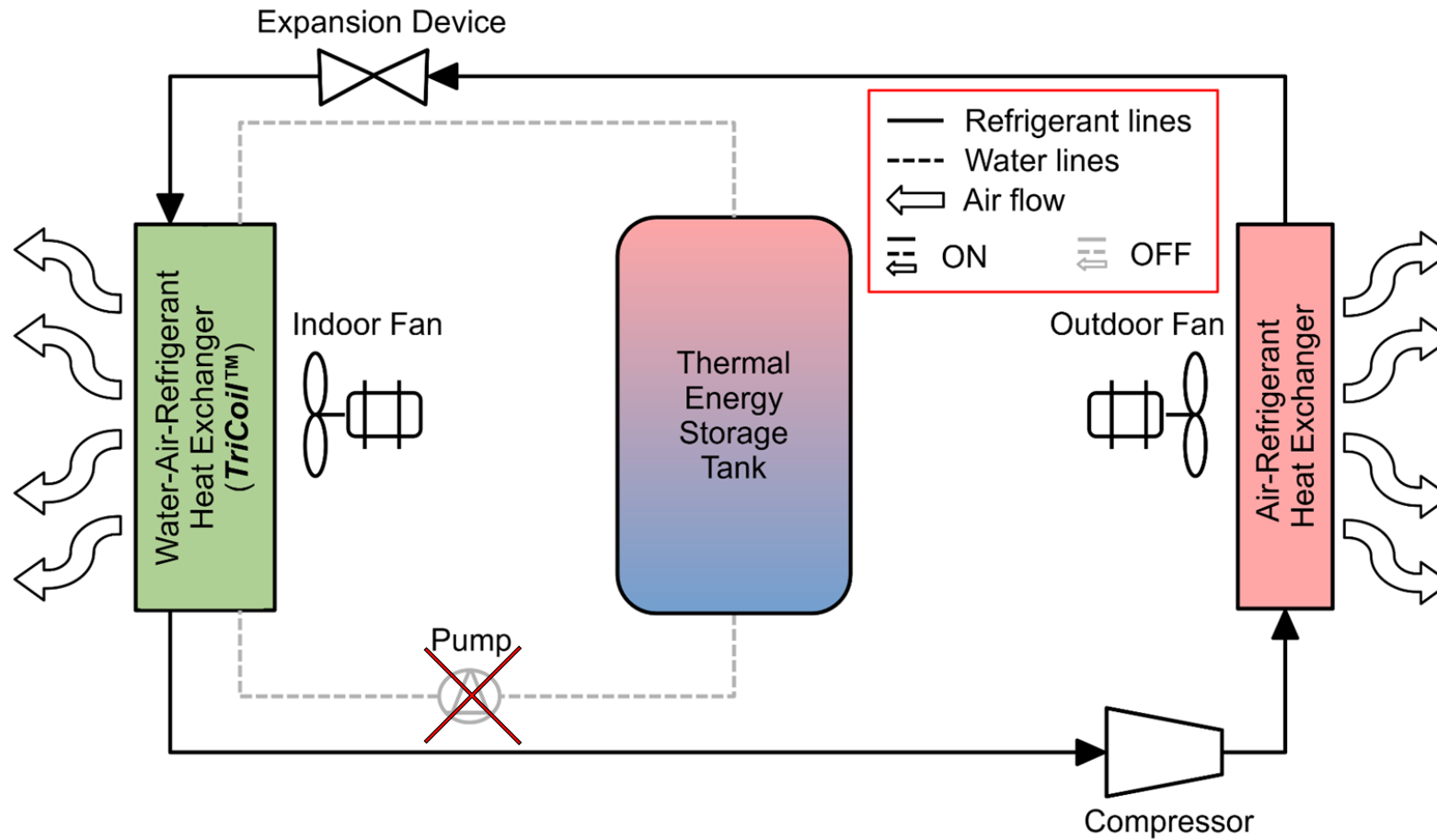


Hydronic System

TriCoil™ in discharging mode acts similar to a hydronic system.



TriCoil™ in direct mode acts similar to a direct expansion system: only 1 heat transfer process on cold side of system!



5 minute Q&A

Objectives

- ✿ Have students ask for clarification/identify gaps**
- ✿ Ask questions/foster discussions if no questions from audience**

15 minutes: Thermal energy storage MPC

Objectives

- ⚙ General intro to TES controls (charging time/duration)**
- ⚙ MPC general principle (objectives, optimization horizon, etc.)**
- ⚙ MPC impact for WTES**

Thermal energy storage – need to make some decisions how to design and run the system!

✿ Conventional Cooling system:

- Track a setpoint (*on/off* or *PI(D)* control)

✿ TES tank size

- Larger tank = more storage capacity
- Larger tank = more cost, more heat gains/losses

✿ TES based system, AC mode, generally:

- Charge when cheap electricity, low carbon emissions, and/or low temperature
- Discharge when electricity is costly
- Cover rest of time in direct mode

✿ TES system controls, questions:

- How much to charge the tank?
- When to charge the tank?
- When to discharge the tank?

} Optimization problem,
need models to do
things 'best'.

Thermal energy storage – rules-based control algorithm for AC is (fairly) simple.

⚙ **How much to charge the tank?**

- Fully, e.g. to a setpoint

⚙ **When to charge the tank?**

- When electricity cost and temperature are lowest: early morning

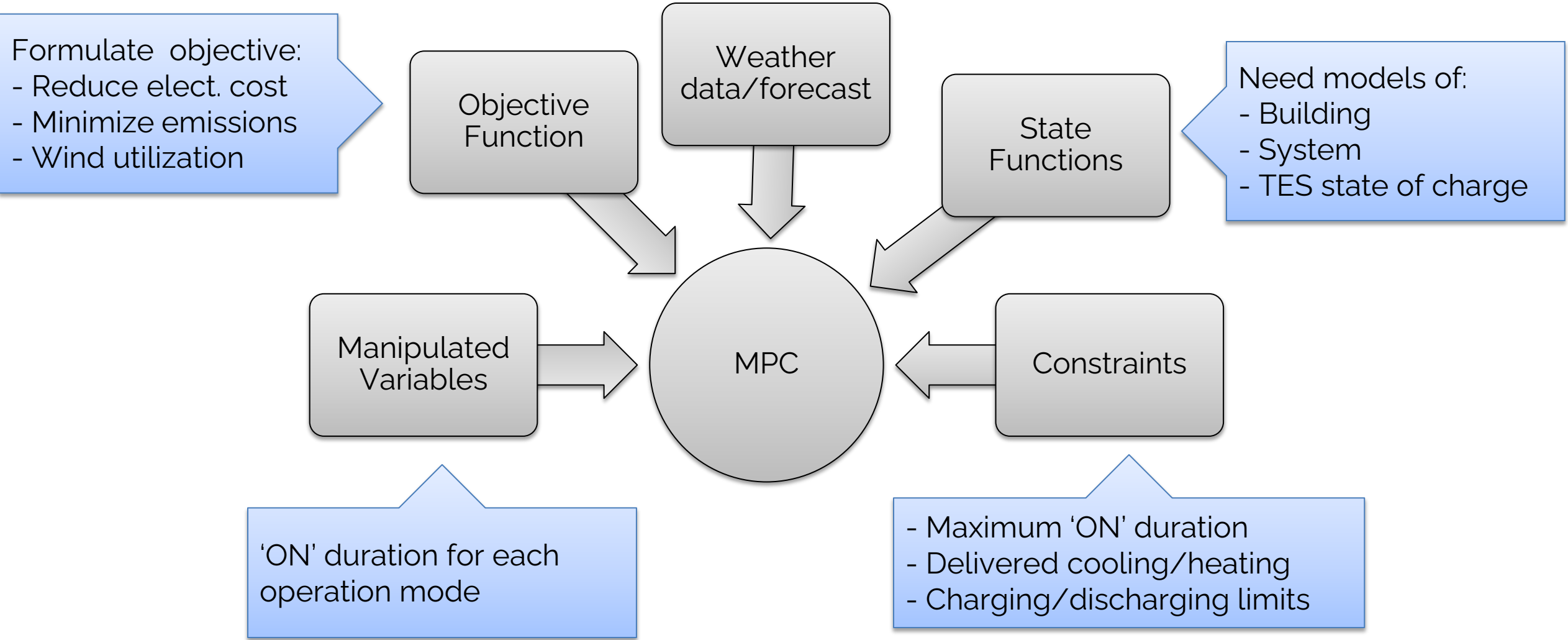
⚙ **When to discharge the tank?**

- When electricity cost and temperature are highest: early afternoon

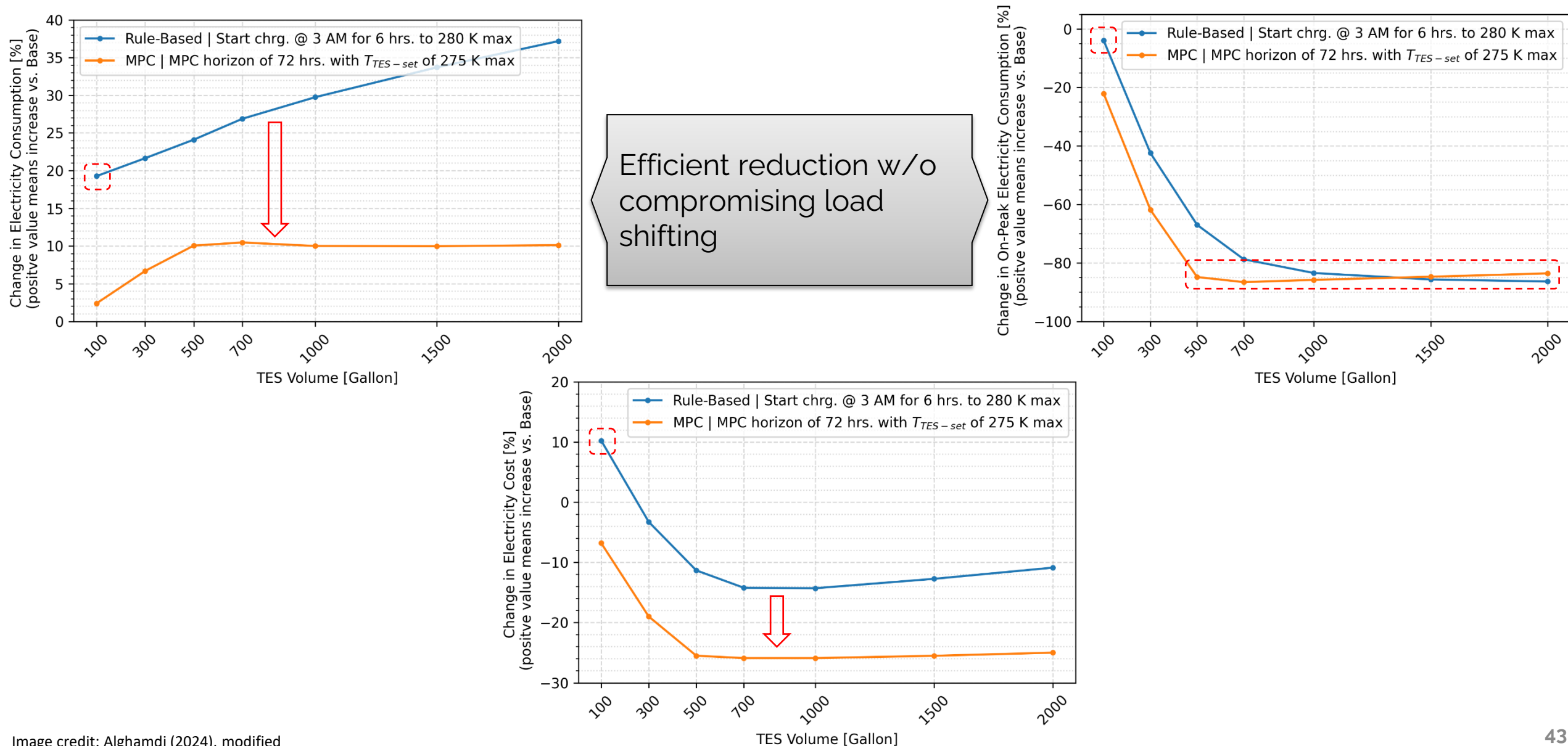
⚙ **What's wrong with the above?**

- A fully charged tank that is not needed gains heat
- Does not respond to changes in ambient conditions

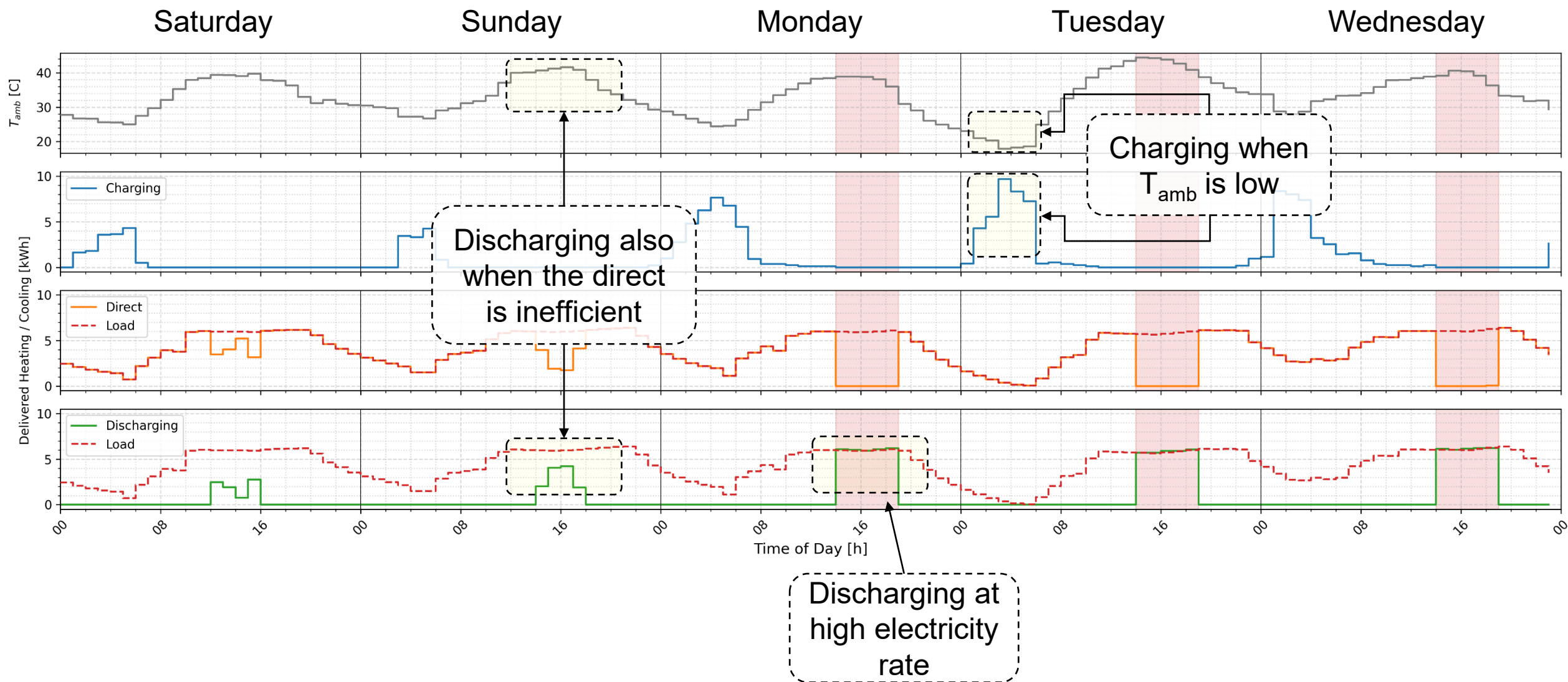
Model predictive control is more complex to set up!



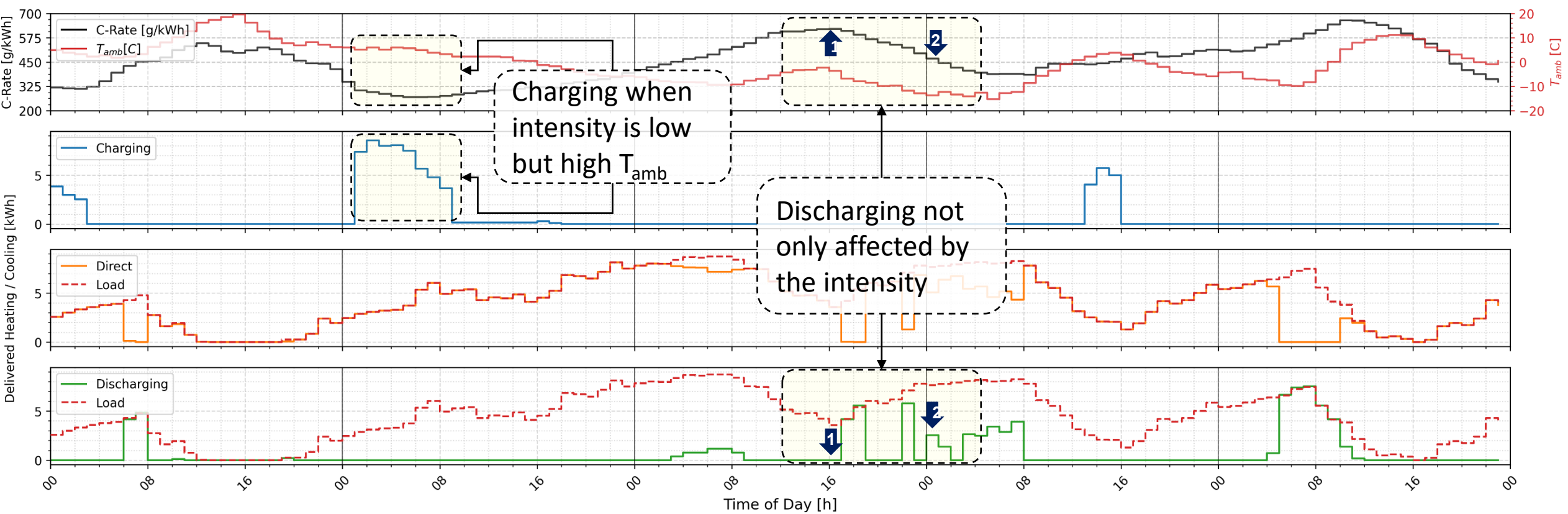
MPC outperforms rules-based controls on minimizing costs!



MPC shifts electricity consumption around to minimize costs.

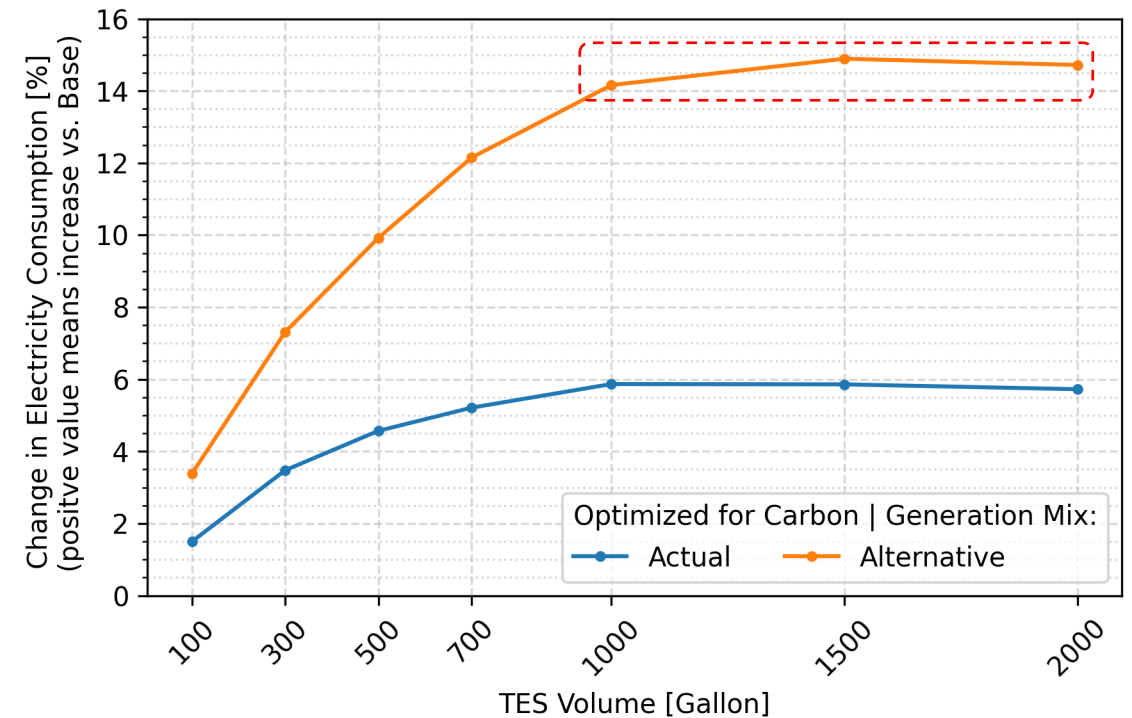
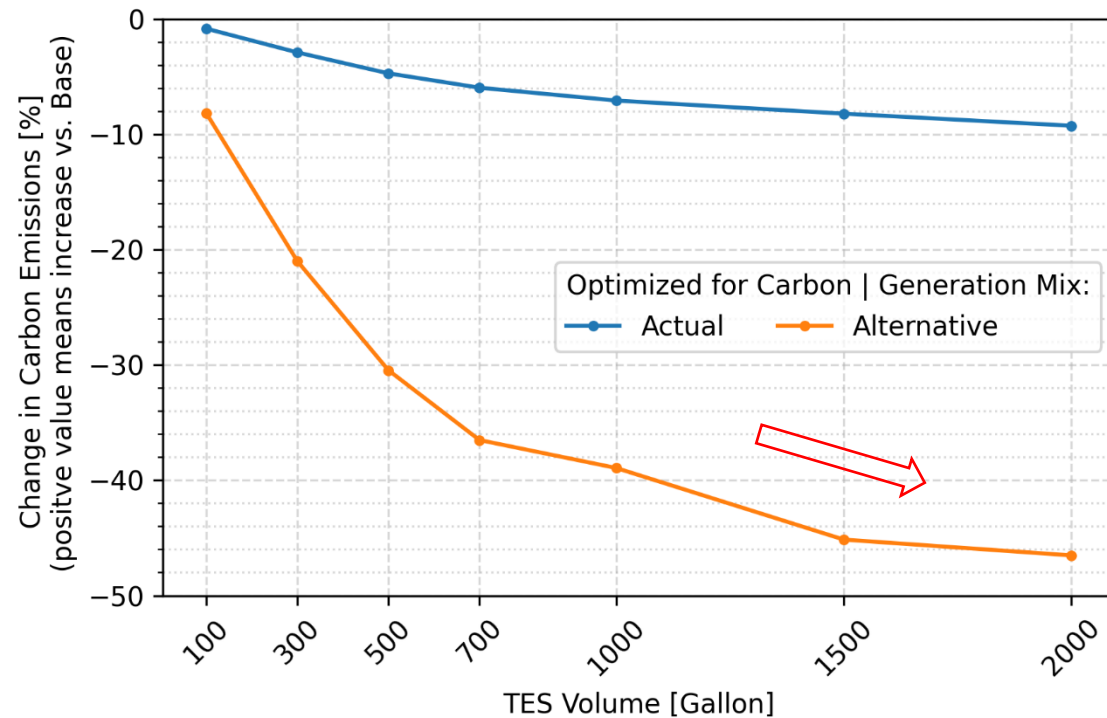


MPC can also be set to reduce carbon emissions!



Carbon savings depend on tank size and electricity mix, likewise, does electricity consumption!

- ✿ Alternative mix with higher renewable content has increase relative reduction potential
- ✿ Increase in tank size has diminishing returns



Summary and Outlook for Thermal Energy Storage

- ✿ **Allows shifting of loads**
- ✿ **Complex systems, even more complex controls**
- ✿ **Requirement of MPC for (near) optimal design and operation**
- ✿ **What else?**
 - Effect of weather data uncertainty?
 - Effect of discrepancy between model and “as installed”?
 - Effect of transient behavior?
 - Effect of variable speed operation?

5 minute Q&A

Objectives

- ⌘ Have students ask for clarification/identify gaps**
- ⌘ Ask questions/foster discussions if no questions from audience**

We covered the combination of electrical supply and use for buildings today!

- ⌘ Electrical power: Supply-side, Demand-side**
- ⌘ Heating and cooling systems – Intro**
- ⌘ Breakout room with exercise (home energy guesstimate)**
- ⌘ Heat pumps, water-based energy storage and its controls**

More questions, if you'd like to dream up a better world.

⚙ **Aditya:** People – Planet – Profit

- Do all people profit, or do the systems we set up primarily profit an exclusive minority?
- Do we profit in the short term or in the long term?
- Do we take debt by causing a high societal cost of carbon emissions (and other pollution) down the road?
- What \$ amount do we give a planet that allows us to exist?

⚙ **We consider individual(s) wealth, beauty, and intelligence as measures of success.**

- Does the above help us in the long-run as a now global society?
- Should we move to a different value set?
- Are we willing to move away from prejudices based on superficial measures towards a society that values all life on the planet?

Outlook – learn about designing buildings that are great for people!

- ⌘ Design of Low-Impact Occupant Centric Buildings**
- ⌘ Examples of Existing Buildings**
- ⌘ Integration of Architecture and Renewable Energy**

References

CAISO (2016). *Fun facts: What the duck curve tells us about managing a green grid*. California Independent System Operator.

Alghamdi, K. (2024). *Development of Component and System Models for a Novel Residential Vapor-Compression System Integrating Water-Based Thermal Energy Storage and Incorporating Both Rule-Based and Model Predictive Control Options*. PhD Defense Slides. Oklahoma State University.

MacCracken, M. (2010). *Energy Storage: Providing for a Low-Carbon Future*. ASHRAE Journal. September 2010.

NBI (2019). *Roadmapping New York Buildings in a Decarbonizing Electric Power Sector*. Presentation at NESEA 2019.

Additional references included as links in slides.